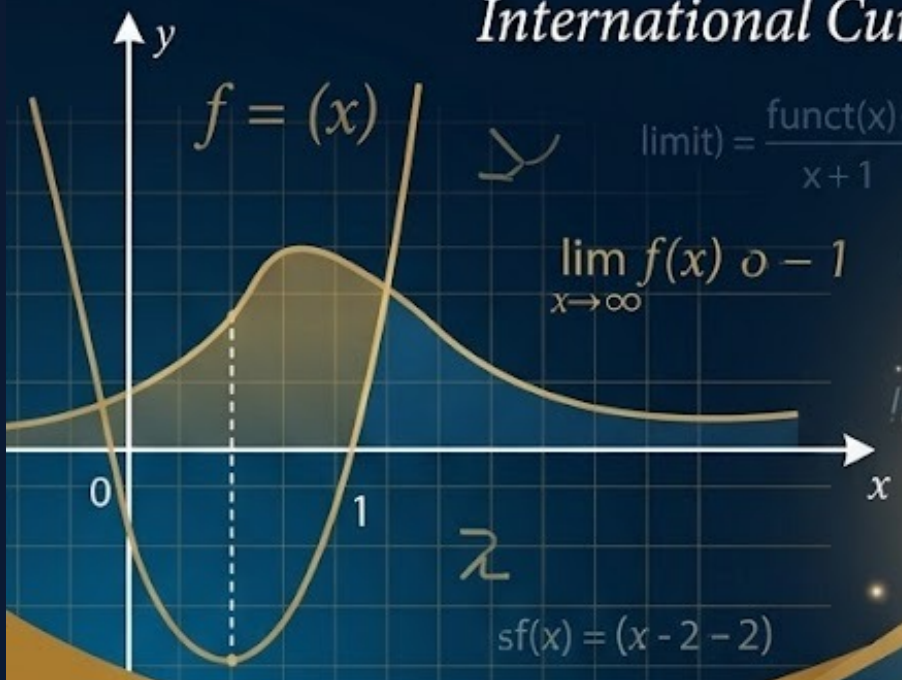
UNITED
INTERNATIONAL ACADEMY
Shaping Global Minds.

بري كالكولاس
PRE-CALCULUS

منهاج الـ AP و الدولي
International Curriculum





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NUMBER #1

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AP Precalc

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1.1 Change in Tandem

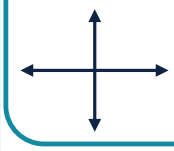
1.1 Notes

Function

A **function** is a mathematical relation that maps a set of _____ values to a set of _____ values such that each input value is mapped to exactly ___ output value.

Input values = = variable

Output values = = variable



Often, it is useful to use other variables that help us recognize the representation. For example, if we throw a football and measure the height, we could use variables where is the height of the football and is the number of seconds since the football was thrown.

Function notation helps us immediately identify which variables are the input and which are the output. For the football example, we would use…

1. $w(x)$ is a function where w is the amount of water in a pool measured in gallons, and x is the length of the pool.
   a. What does the independent variable represent?

   b. What does the dependent variable represent?

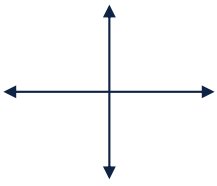
These two variables will **vary in tandem** according to the function rule that is established. That rule can be expressed **V**erbally, **A**nalytically, **N**umerically, or **G**raphically… . Throughout this course, you will use all four of these types of expressions to represent a function.

Increasing function if…

Verbally: …as the input values increase, the output values always

Analytically: …for all a and b in the interval, if

Graphically:



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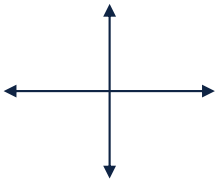
Write your questions and thoughts here!

Decreasing function if…

Verbally: …as the input values increase, the output values always

Analytically: …for all a and b in the interval, if

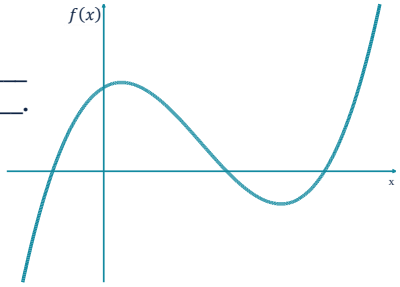
Graphically:



2. Example of **numerically**: Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the interval and **justify**.

x	4	6	7	10	20
$f(x)$	1	1.01	1.04	1.06	1.29

Basic elements of a function's graph



- Zero

The graph intersects the x-axis (independent) when the _____ value is zero. We call the INPUT values _____.

- y-intercept

The graph intersects the y-axis (dependent) when the _____ value is zero. We call the output value the _____.

- Concavity.

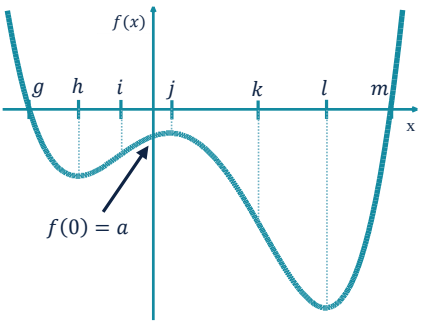
  Concave UP

  Concave DOWN

  Straight lines don't have concavity.

  Point of inflection (point where the concavity changes)

Use the graph of f below to answer the following questions.



a. When is the graph concave up?

b. When is the graph concave down?

c. Find the zero(s) of the function.

d. Find the y-intercept(s) of the function.

e. When is the graph increasing?

f. When is the graph decreasing?

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1.1 Change in Tandem

AP Precalculus

1.1 Practice

For each function, identify what the dependent and independent variables represent.

1. $b(s)$ is a function where b is the number of books in the library and s is the number of students in the school.

Dep:

Indep:

2. $p(t)$ is a function where t is the number of years since kindergarten and p is the number of Pokemon cards.

Dep:

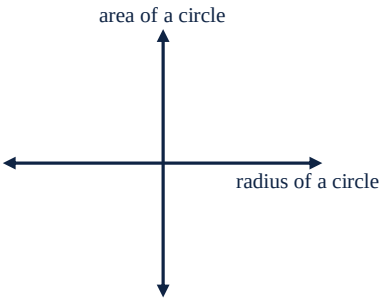
Indep:

3. $c(t)$ is a function where c is the number of cups of coffee consumed and t is the number of teachers at the school.

Dep:

Indep:

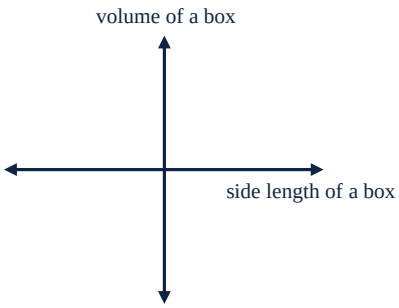
4.



Dep:

Indep:

5.



Dep:

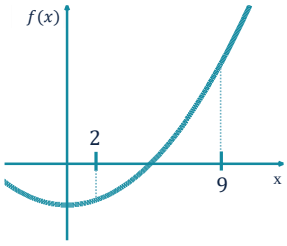
Indep:

Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the given interval and justify your answer.

6.

x	1	2	3	4	5
$f(x)$	95	90	75	50	10

7.

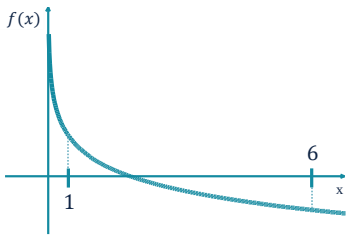


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8.

x	12	16	18	25	31
$f(x)$	8	7.99	7.8	7.75	7.7

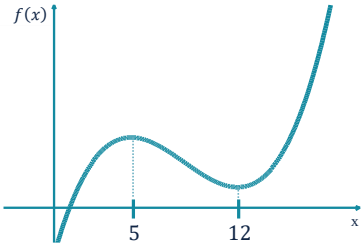
9.



10.

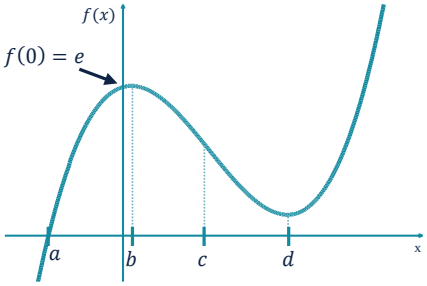
x	101	201	301	401	501
$f(x)$	10	11	12	13	14

11.



Use the graph of f to answer the question below.

12.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

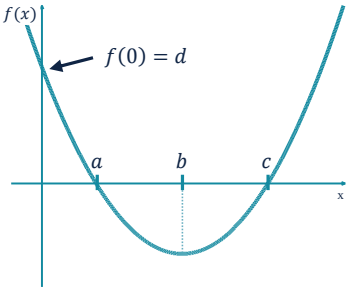
c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

13.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

c. On what interval(s) is the graph increasing?

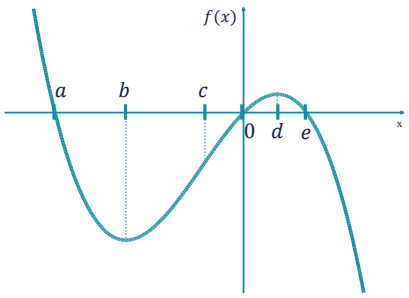
d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

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14.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

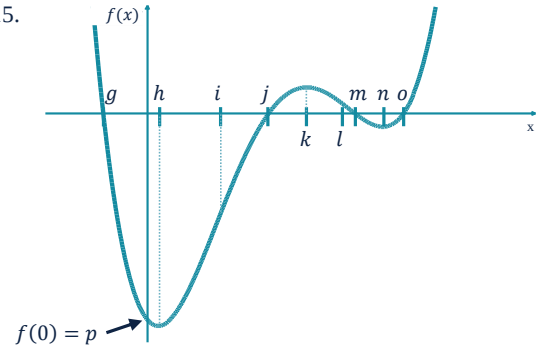
c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

15.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

1.1 Change in Tandem

1.1 Test Prep

Multiple Choice

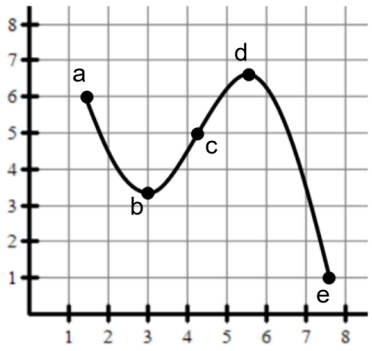
16. Let the function f be defined as a function that is always decreasing. The table shows values for the function f at selected values of x. Which of the following represents all possible values for a ?

(A) $-5 < a < 7$
(B) $-2 < a < 0$
(C) $0 < a < 3$
(D) $-2 < a < 4$
(E) $-15 < a < 12$

x	$f(x)$
-5	12
-2	4
0	a
3	-2
7	-15

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For questions 17-18 use the graph of f.



graph of f

17. The figure shows the graph of a function f . Let a, b, c, d, and e represent the x-coordinates at those points. Of the following, which is a point of inflection?

(A) point a

(B) point b

(C) point c

(D) point d

(E) point e

18. The figure shows the graph of a function f . Let a, b, c, d, and e represent the x-coordinates at those points. Of the following, on which interval is f decreasing and the graph of f concave down?

(A) The interval from a to b.

(B) The interval from b to c.

(C) The interval from c to d.

(D) The interval from d to e.

(E) None of the above.

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1.1 Change in Tandem

AP Precalculus

Name: _____ **CA #1**

For each function, identify what the dependent and independent variables represent.

1. $b(h)$ is a function where h is the initial height of a dropped basketball and b is the height of the first bounce of a basketball.

Dep:

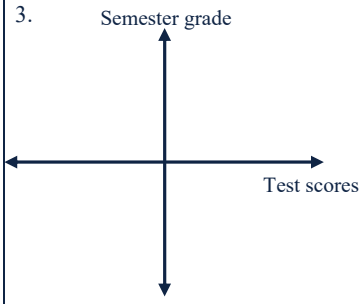
Indep:

2. $h(t)$ is a function where h is the number of hotels in Jamaica and t is the number of tourists to visit Jamaica each year.

Dep:

Indep:

3.



Dep:

Indep:

4. Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the given interval and justify your answer.

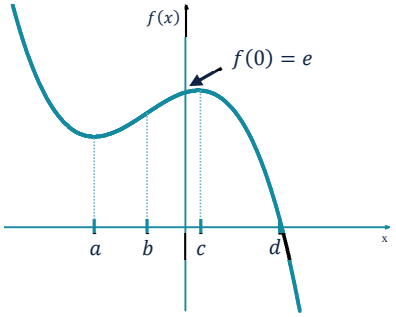
x	7	9	12	16	25
$f(x)$	2	1.95	1.91	1.88	1.87

More on the back.

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Use the graph of f to answer the question below.

5.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

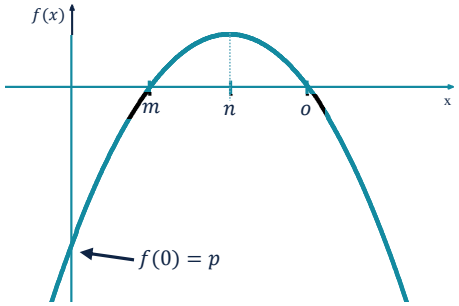
c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

6.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

Answers to 1.1 CA #1

1. **Dep**: Height of the first bounce. **Ind**: Initial height of the basketball.	2. **Dep**: Number of hotels in Jamaica **Ind**: Number of tourists each year.	3. **Dep**: Semester grade **Ind**: Test scores	4. f is decreasing on the interval $7 < x < 25$ because for all a and b in the interval, if $a < b$, then $f(a) > f(b)$.

5a. $x < b$ 5b. $x > b$ 5c. $a < x < c$ 5d. $x < a$ and $x > c$ 5e. $x = d$ 5f. $y = e$	6a. none 6b. $\mathbb{R}$ or $-\infty < x < \infty$ 6c. $x < n$ 6d. $x > n$ 6e. $x = m$ and $x = o$ 6f. $y = p$

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1.1 Change in Tandem

AP Precalculus

Name: _____ **CA #2**

For each function, identify what the dependent and independent variables represent.

1. $b(c)$ is a function where b is the number of billboards on the street and c is the number of cars registered in the city.

Dep:

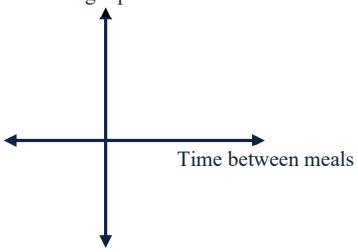
Indep:

2. $t(g)$ is a function where g is the number of girls in the scout troop and t is the number of Thin Mints sold.

Dep:

Indep:

3. Hunger pains



Dep:

Indep:

4. Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the given interval and justify your answer.

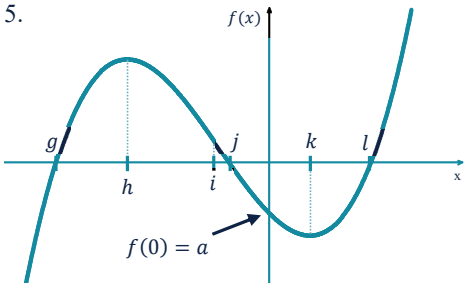
x	3	4	5	6	7
$f(x)$	6	8	10	11	11.5

More on the back.

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Use the graph of f to answer the question below.

5.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

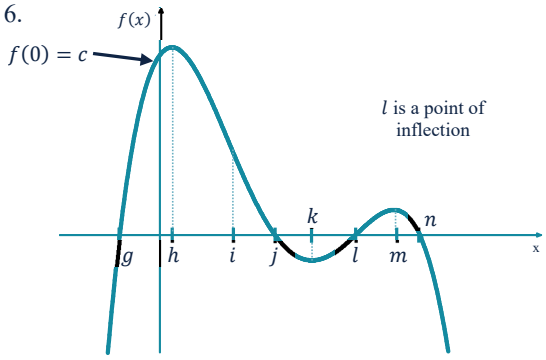
c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

6.



a. On what interval(s) is the graph concave up?

b. On what interval(s) is the graph concave down?

c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.

Answers to 1.1 CA #2

1. **Dep**: Number of billboards. **Ind**: Number of cars registered.	2. **Dep**: Number of Thin Mints sold. **Ind**: Number of girls in the scout troop.	3. **Dep**: Hunger pains **Ind**: Time between meals	4. f is increasing on the interval $3 < x < 7$ because for all a and b in the interval, if $a < b$, then $f(a) < f(b)$.

5a. $x > i$ 5b. $x < i$ 5c. $x < h$ and $x > k$ 5d. $h < x < k$ 5e. $x = g, x = j$ and $x = l$ 5f. $y = a$	6a. $i < x < l$ 6b. $x < i$ and $x > l$ 6c. $x < h$ and $k < x < m$ 6d. $h < x < k$ and $x > m$ 6e. $x = g, x = j, x = l$, and $x = n$ 6f. $y = c$

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AP Precalc

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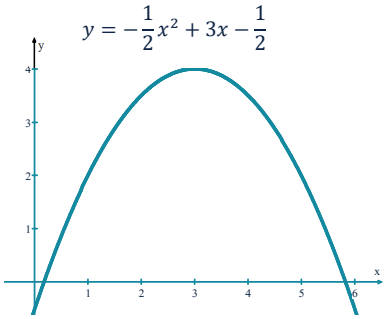
1.2 Rates of Change

1.2 Notes

Average Rate of Change

_____ = ____ = _____ =

1. Find the average rate of change between $(1, 2)$ and $(3, 4)$.



Instead of a graph, let's say we have a table of values, and you want to calculate the average rate of change on an interval. Be careful that the **output values are always in the numerator**.

2. What is the average rate of change on the interval from 13 to 25 seconds?

t time (seconds)	10	13	19	25
$d(t)$ distance (meters)	80	76	43	30

The word ***per*** is a key indicator that you are dealing with a rate of change. Miles ***per*** gallon. Students ***per*** classroom. Online gamers **per** server. It also tells you which variable is dependent or independent, as the dependent is always listed first.

Rate of Change at a Point

What about the rate of change of a function at a point? This tells us about the rate at which the output values would change were the input values to change at that point. We can approximate the rate of change at a point by using the average rates of change over small intervals that contain the point.

3. Estimate the rate of change at $x = 1$ for the function $f(x) = -\frac{1}{2}x^2 + 3x - \frac{1}{2}$.

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Write your questions and thoughts here!

4. Try it on your own! Find an approximation for the rate of change of g at $x = -2$ if $g(x) = e^x$.

Positive and Negative Rates of Change

A **positive** rate of change indicates that as one quantity increases or decreases, the other quantity _____.

A **negative** rate of change indicates that as one quantity increases, the other _____.

State whether the situation represents a positive or negative rate of change.

5. As the years increase, a high school student body increases.

6. As Mr. Bean's weight decreases, his running distances increase.

1.2 Rates of Change

AP Precalculus

1.2 Practice

Find the average rate of change of the function on the given interval.

1. $h(t) = 3t - t^2$ over the interval $2 \le t \le 5$.

2. $b(w) = w + 2^w$ over the interval $[-1, 2]$.

3. $f(x) = \ln 3x$ over the interval $1 \le x \le 4$.

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Use the information in each table to find the average rate of change on the given interval.

4.

t Minutes	3	10	21	43	55
$d(t)$ meters	5	102	135	140	143

a. $3 \leq t \leq 55$

b. $10 \leq t \leq 43$

c. $21 \leq t \leq 55$

5.

t months	6	12	24	48	72
$d(t)$ hair follicles	20,000	19,800	15,000	10,000	7,500

a. $6 \leq t \leq 48$

b. $12 \leq t \leq 72$

c. $6 \leq t \leq 72$

Estimate the rate of change of each function at the given point.

6. $f(x) = \frac{1}{3x}$ at $x = 4$

7. $f(x) = 2x^2 + 1$ at $x = -2$

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8. $f(x) = 7\sqrt{x}$ at $x = 2$

9. $f(x) = \ln(2x)$ at $x = 3$

State whether the situation represents a positive or negative rate of change.

10. A candy company uses pints of chocolate to make candy. The more chocolate they use, the more boxes of candy are produced.

11. The amount of money is Josh's savings account decreases for each semester he attends college.

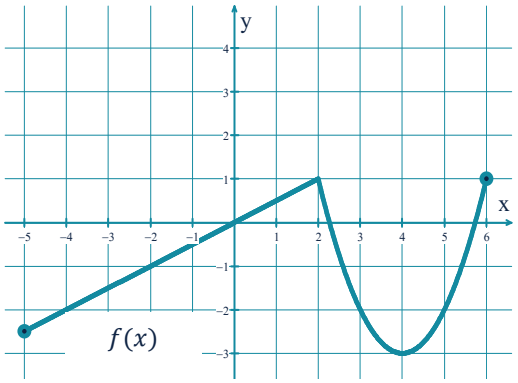
12. As the number of cats Mr. Sullivan owns increases, the number of mice in his barn decreases.

13. As the amount of water Mr. Brust drinks decreases, the fewer trips to the restroom he needs to make.

1.2 Rates of Change

1.2 Test Prep

14. A continuous function f is defined on the closed interval $-5 < x < 6$ and is shown in the graph below. For how many values of b, $-5 < b < 6$, is the average rate of change of f on the interval $[b, 5]$ equal to 0? Give a reason for your answer.



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1.2 Rates of Change

CA #1

AP Precalculus Name: _____

Find the average rate of change of the function on the given interval.

1. $f(x) = \sqrt{5x - 2}$ on the interval $0.5 \leq x \leq 3$.

2. $w(t) = 5t^2 - 5t + 1$ on the interval $[-2, 1]$

Use the information in the table to find the average rate of change on the given interval.

3.

t minutes	1	3	4	12	13
$s(t)$ feet	-2	4	-7	5	10

a. $3 \leq t \leq 13$

b. $1 \leq t \leq 12$

c. $3 \leq t \leq 4$

Estimate the rate of change of each function at the given point.

4. $f(x) = \frac{3}{x}$ at $x = 4$

5. $f(x) = 4x^2 - 1$ at $x = 2$

6. $f(x) = \sqrt{5x}$ at $x = 1$

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State whether the situation represents a positive or negative rate of change.

7. As you do more chores, your parents' happiness level increases.

8. The electrostatic force, measured in newtons, between two charged particles decreases as the distance between the particles increases.

9. As the temperature decreases, the number of people at the lake decreases.

Answers to 1.2 CA #1

1. 1.159	2. -10	3a. 0.6 feet / min 3b. 0.636 feet / min 3c. -11 feet / min	4. ≈ -0.1875	5. ≈ 16
6. ≈ 1.118	7. positive	8. negative	9. positive	

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1.2 Rates of Change

CA #2

AP Precalculus Name: _____

Find the average rate of change of the function on the given interval.

1. $f(x) = \ln(7x)$ on the interval $3 \le x \le 10$.

2. $s(t) = \frac{1}{t-3}$ on the interval $[0, 1]$

Use the information in the table to find the average rate of change on the given interval.

3.

h horses	1	5	12	20
$f(h)$ flies	20	74	205	516

a. $1 \le t \le 20$

b. $5 \le t \le 12$

c. $5 \le t \le 20$

Estimate the rate of change of each function at the given point.

4. $f(x) = 4x - x^2 + 2$ at $x = -2$

5. $f(x) = \frac{1}{2x}$ at $x = -2$

6. $f(x) = \sqrt{x}$ at $x = 5$

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State whether the situation represents a positive or negative rate of change.

7. The number of logs on the campfire increases and the temperature of the fire also increases.

8. As the miles you drive increases, the amount of gas in the tank decreases.

9. The number of pass attempts by a quarterback decrease and his number of interceptions also decreases.

Answers to 1.2 CA #2

1. 0.1719	2. $-\frac{1}{6} \approx -0.1666$	3a. 26.105 flies per horse 3b. 26.4285 flies per horse 3c. 29.4666 flies per horse		4. ≈ 8
5. ≈ -0.125	6. ≈ 0.2236	7. positive	8. negative	9. positive

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AP Precalc

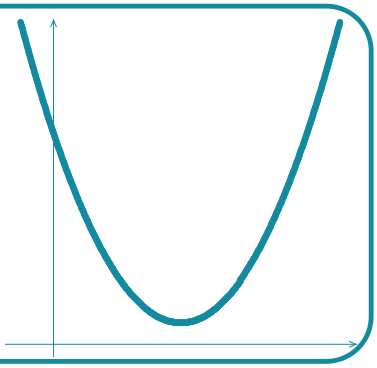
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1.3 Rates of Change in Linear and Quadratic Functions

1.3 Notes

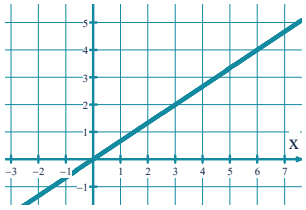
Avg Rate of Change = Slope of a Secant Line

The average rate of change over the closed interval $[a, b]$ is the slope of the secant line from the point $(a, f(a))$ to $(b, f(b))$.

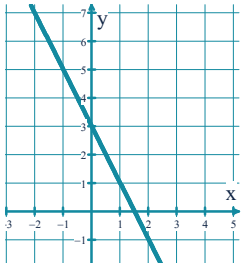


Find the average rate of change for each linear function.

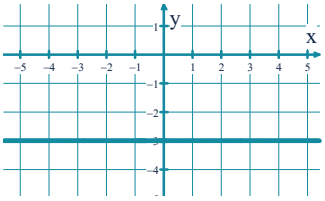
1. $y = \frac{2}{3}x$



2. $y = -2x + 3$



3. $y = -3$



The average rate of change for a linear function is _____. Regardless of the input-value interval length, the average rate of change stays the same!

Find the rate of change of the average rate of change of each linear function.

4. $y = \frac{2}{3}x$

5. $y = -2x + 3$

6. $y = -3$

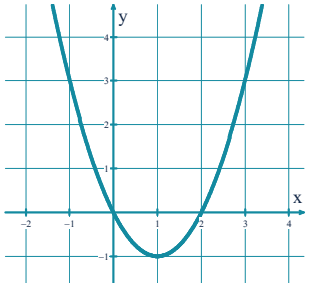
The rate of change of the *average* rates of change of a linear function is ____.

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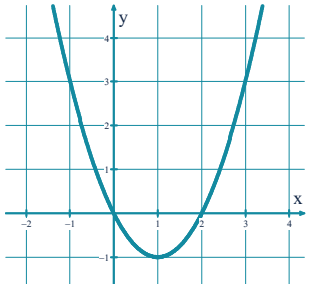
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Find the average rate of change for the quadratic function $y = x^2 - 2x$ on the given interval.

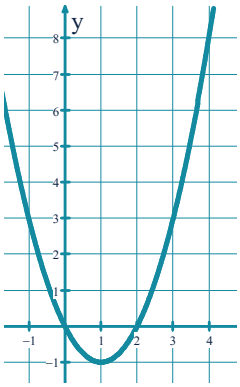
7. $1 \le x \le 2$



8. $2 \le x \le 3$



9. $3 \le x \le 4$



10. $[4, 5]$

11. $[5, 6]$

The average rate of change for a quadratic function does not stay the same. For consecutive equal-length input-value intervals, the rate of change of the average rates of change of a quadratic function is _____. For the above problem, this constant is _____.

12. $f(x) = -2x^2 + 10$.

13. $f(x) = 3x^2 - 2x$.

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14. The values of a function g are given at selected x-values in the table below. The function is either concave up or concave down, but not both.

x	15	16	17	18	19
$g(x)$	18	20	20	18	14

Determine if the function is concave up or concave down.

1.3 Rates of Change in Linear and Quadratic Functions

AP Precalculus

1.3 Practice

What is the average rate of change for each function on the given intervals?

1. $y = 7x + 10$ on $-6 \leq x \leq -4$

2. $y = 5x - 3$ on $1 \leq x \leq 7$

3. $y = x^2 + 4x - 2$ on $-3 \leq x \leq 2$

4. $y = 2x^2 + 2x + 1$ on $2 \leq x \leq 5$

For each problem, a quadratic function g has the given average rates of change on an interval.

5. Avg rate of change is 1 on the interval $-3 \leq x \leq 0$.
Avg rate of change is 5 on the interval $0 \leq x \leq 3$.
What is the average rate of change of g on the interval $3 \leq x \leq 6$?

6. Avg rate of change is -1 on the interval $1 \leq x \leq 5$.
Avg rate of change is -7 on the interval $5 \leq x \leq 9$.
What is the average rate of change of g on the interval $17 \leq x \leq 21$?

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7. Avg rate of change is 2 on the interval $-1 \le x \le 1$.
Avg rate of change is 7 on the interval $1 \le x \le 3$.
What is the average rate of change of g on the interval $-3 \le x \le -1$?

8. Avg rate of change is -6 on the interval $0 \le x \le 5$.
Avg rate of change is -2 on the interval $5 \le x \le 10$.
What is the average rate of change of g on the interval $20 \le x \le 25$?

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

9. $y = 13x - 10$

10. $f(x) = x^2 + 3x + 8$

11. $f(x) = 6x - 2x^2 + 1$

The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

12.

x	-8	-6	-4	-2	0
$g(x)$	10	11	14	20	29

13.

x	5	10	15	20	25
$h(x)$	100	75	40	0	-50

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1.3 Rates of Change in Linear and Quadratic Functions

1.3 Test Prep

14. The table gives values of the function h for selected values of x.

x	-8	-4	4
$h(x)$	23	17	5

If the function h is linear, what is the value of $h(15)$?

(A) $-\frac{21}{2}$

(B) $-\frac{23}{2}$

(C) -5

(D) -11

15. The table gives values of the function g for selected values of x.

x	0	2	4
$g(x)$	-5	-1	0

If the function g is quadratic, what is the value of $g(8)$?

(A) -5

(B) 5

(C) -7

(D) 7

16. The function p is given by $p(x) = g(x+1) - g(x)$. If $p(x) = 2$, which of the following statements must be true?

I. Because p is positive and constant, the graph of g always has positive slope.
II. Because p is positive and constant, the graph g is concave up.
III. Because p is positive and constant, g is increasing

(A) I only (B) II only (C) III only (D) I and III only

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1.3 Rates of Change in Linear and Quadratic Functions

AP Precalculus

Name: _____ **CA #1**

What is the average rate of change for each function on the given intervals?

1. $y = 3x - 6$ on $-3 \leq x \leq 2$

2. $y = -8 - 2x$ on $5 \leq x \leq 7$

3. $y = x^2 + 3x - 5$ on $-2 \leq x \leq 4$

4. $y = 2x^2 - 6x + 4$ on $-3 \leq x \leq -2$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

5. $y = 21x - 1$

6. $y = 8 - x$

7. $f(x) = -x^2 + 3x - 7.$

8. $f(x) = 3x^2 + x - 1.$

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The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

9.

x	4	7	10	13	16
$g(x)$	35	20	13	10	9

10.

x	-5	-3	-1	1	3
$h(x)$	100	95	80	50	10

Answers to 1.3 CA #1

1. 3	2. -2	3. 5	4. -16	5. 0	6. 0	7. -2
8. 6	9. Concave up because the rate of change is increasing over equal-length input-value intervals.			10. Concave down because the rate of change is decreasing over equal-length input-value intervals.		

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1.3 Rates of Change in Linear and Quadratic Functions

AP Precalculus

Name: _____ **CA #2**

What is the average rate of change for each function on the given intervals?

1. $y = 4x - 1$ on $-10 \leq x \leq -5$

2. $y = 8 - 3x$ on $-3 \leq x \leq 5$

3. $y = x^2 - 4x - 6$ on $-1 \leq x \leq 4$

4. $y = -2x^2 + 3x - 2$ on $2 \leq x \leq 3$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

5. $y = 5 + 8x$

6. $y = -3x + 10$

7. $f(x) = 5x - 4x^2$

8. $f(x) = 3x^2 + x - 2$

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The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

9.

x	-12	-7	-2	3	8
$g(x)$	-10	-12	-16	-22	-32

10.

x	1	2	3	4	5
$h(x)$	10	-10	-22	-27	-28

Answers to 1.3 CA #2

1. 4	2. -3	3. -1	4. -7	5. 0	6. 0	7. -8
8. 6	9. Concave down because the rate of change is decreasing over equal-length input-value intervals.			10. Concave up because the rate of change is increasing over equal-length input-value intervals.		

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AP Precalc

Write your questions and thoughts here!

1.4 Polynomial Functions and Rates of Change

1.4 Notes

Polynomial

"Poly" means many, or much. A nonconstant polynomial is a function with many terms in the following form:

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0$$

where n is a positive integer and a_i is a real number for each i from 1 to n.

Quick example: $p(x) =$

The leading term is _____ | The polynomial has degree ____ | The leading coefficient is _____

1. Example from above.
   What is the degree of the polynomial? What is the leading coefficient?

Sometimes, the polynomial will **not** be written in standard form. You have to look at the variables LARGEST exponent to find the leading term.

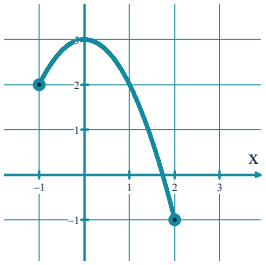
2. $f(x) = 14x^2 + 6x - 2x^6 + 1$
   What is the degree of the polynomial? What is the leading coefficient?

Local (Relative) Extrema (maxima and minima)

If a polynomial switches from increasing to decreasing, there will be a _____, or _____, _____ output value.

If a polynomial switches from decreasing to increasing, there will be a _____, or _____, _____ output value.

An **included endpoint** of a polynomial with a restricted domain may also have a local, or relative, extrema.



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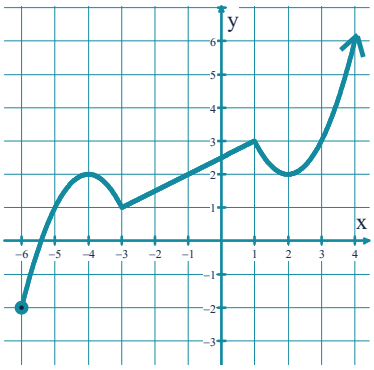
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Global (Absolute) Extrema (maxima and minima)

Of all local maxima, the greatest is called the **global, or absolute**, maximum.

Of all local minima, the least is called the **global, or absolute**, minimum.

Find the extrema and where they occur. If there are none, cross it off and write NONE.



3. Absolute **min** of _____ when $x =$

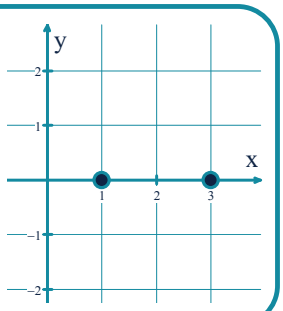
4. Absolute **max** of _____ when $x =$

5. Relative **min**(s) at $x =$

6. Relative **max**(es) at $x =$

Two Zeros = at least one extremum

If you have _____ of a polynomial (nonconstant polynomial), then there must be at least _____ local extremum between the two zeros.



7. Let f be a polynomial function with values $f(-2) = 0, f(0) = 5, f(4) = 0,$ and $f(7) = -1$. Are there any guaranteed extrema? If so, state where they occur.

Even Degree = Absolute Extrema

If the polynomial has an even degree, there is either an absolute maximum or an absolute minimum.

Positive leading coefficient means you will have an absolute _____

Negative leading coefficient means you will have an absolute _____

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Write your questions and thoughts here!

8. Is there an absolute maximum or minimum for each function?

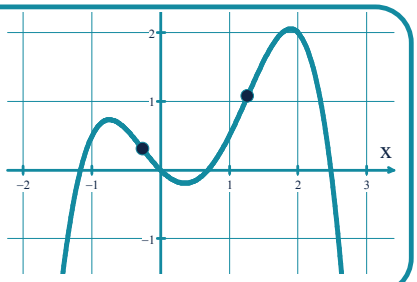
a. $f(x) = 3x^4 - 2x^3 - 5$

b. $f(x) = -x^6 + x^2 - x + 5$

c. $f(x) = 6x^3 + x^2 - 2$

Point of Inflection

A point of inflection occurs at input values where the rate of change changes from increasing to decreasing (or vice versa). This occurs when the graph is changing from concave up to concave down (or vice versa).



1.4 Polynomial Functions and Rates of Change

AP Precalculus

1.4 Practice

Find the leading coefficient and the degree of each polynomial.

1. $f(x) = 8x^4 - 4x^3 + 6x^2 + 10$

L.C. _____ Degree: _____

2. $f(x) = 8x^2 - 3x$

L.C. _____ Degree: _____

3. $f(x) = -5x^7 + 6x^4 - x$

L.C. _____ Degree: _____

4. $f(x) = 3x^4 + 10x^5 - 8x^3 + 1$

L.C. _____ Degree: _____

5. $f(x) = 5x^3 - 9x^2 + x^7 - 3x^8$

L.C. _____ Degree: _____

6. $f(x) = 9x^6 - 2x^7$

L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

7. $f(-1) = 0, f(0) = 6$, and $f(6) = 0$.

8. $f(0) = 6, f(3) = 2, f(6) = 0$, and $f(10) = 0$.

9. $f(-5) = 0, f(0) = 5$, and $f(5) = 7$.

Is there a global maximum or minimum for each function?

10. $f(x) = 2x^8 - x^3 + x^2 + 6$

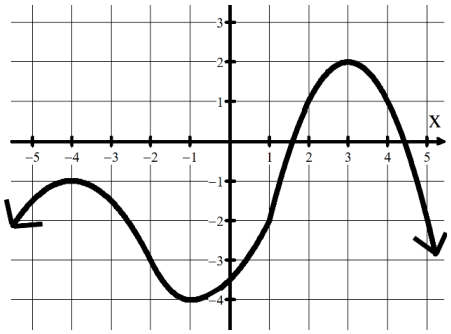
11. $f(x) = -3x^4 + 5x - 1$

12. $f(x) = 6x^7 + 3x^4 - 4x + 2$

13. $f(x) = -4x^6 - 10x^2 - 7$

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Find the following extrema. If there are none, cross it off and write NONE.

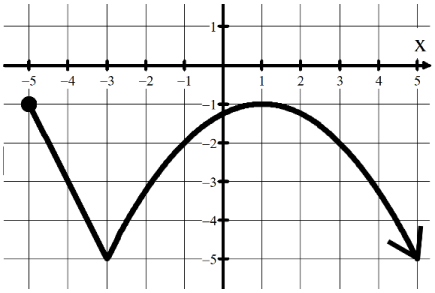


14. Absolute **min** of _____ when $x =$

15. Absolute **max** of _____ when $x =$

16. Relative **min**(s) at $x =$

17. Relative **max**(es) at $x =$

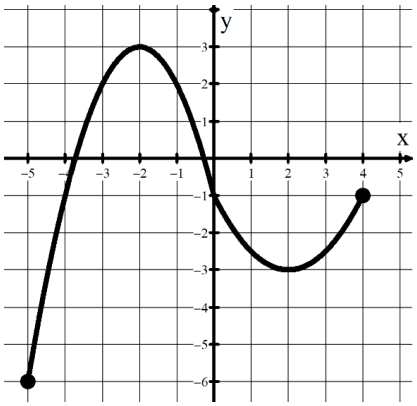


18. Absolute **min** of _____ when $x =$

19. Absolute **max** of _____ when $x =$

20. Relative **min**(s) at $x =$

21. Relative **max**(es) at $x =$



22. Absolute **min** of _____ when $x =$

23. Absolute **max** of _____ when $x =$

24. Relative **min**(s) at $x =$

25. Relative **max**(es) at $x =$

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1.4 Polynomial Functions and Rates of Change

1.4 Test Prep

26. For each of the following polynomials, determine if there is an absolute minimum, absolute maximum, or neither.

a. $f(x) = 3x(x-5)(x+2)^2$

b. $g(x) = -4(x^3+1)(x-3)^2$

c. $h(x) = (x^2-6x+3)(2-x^4)$

27. **Calculator active.** For $1 \le t \le 4$, the number of people waiting in line to ride a new rollercoaster at time t hours can be modeled by the function $L(t) = -0.56t^5 + 2.31t^4 - 0.264t^3 + 1$. Based on this model, at what time t does the number of people waiting in line change from increasing to decreasing?

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1.4 Polynomial Functions and Rates of Change

CA #1

AP Precalculus Name: _____

Find the leading coefficient and the degree of each polynomial.

1. $f(x) = -8x^2 - 6x - 5$

   L.C. _____ Degree: _____

2. $f(x) = 10x^3 + 7$

   L.C. _____ Degree: _____

3. $f(x) = 8x^4 + 3x^5 - 2x$

   L.C. _____ Degree: _____

4. $f(x) = 2x + 8x^4$

   L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

5. $f(0) = -2, f(1) = 0,$ and $f(11) = 0.$

6. $f(-8) = 0, f(0) = 1,$ and $f(5) = 3.$

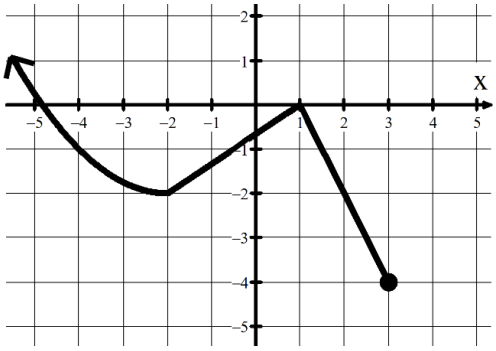
Is there a global maximum or minimum for each function?

7. $f(x) = 2x^5 - 4x^2 + x$

8. $f(x) = 3x^6 - 2$

9. $f(x) = -7x^8 + 2x^3 + 1$

Find the following extrema. If there are none, cross it off and write NONE.



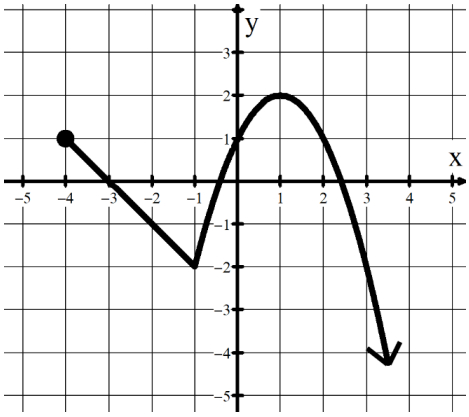
10. Absolute **min** of _____ when $x =$

11. Absolute **max** of _____ when $x =$

12. Relative **min**(s) at $x =$

13. Relative **max**(es) at $x =$

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14. Absolute **min** of _____ when $x =$

15. Absolute **max** of _____ when $x =$

16. Relative **min**(s) at $x =$

17. Relative **max**(es) at $x =$

Answers to 1.4 CA #1

1a. -8 1b. 2	2a. 10 2b. 3	3a. 3 3b. 5	4a. 8 4b. 4	5. Yes, on $1 < x < 11$.
6. No guarantee.	7. none	8. min	9. max	10. min of -4 when $x = 3$
11. none	12. -2 and 3	13. 1	14. none	15. max of 2 when $x = 1$
16. -1			17. -4 and 1	

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1.4 Polynomial Functions and Rates of Change

CA #2

AP Precalculus Name: _____

Find the leading coefficient and the degree of each polynomial.

1. $f(x) = 3x + 1$

L.C. _____ Degree: _____

2. $f(x) = -3x^4 - 10x^3 - 4x + 3$

L.C. _____ Degree: _____

3. $f(x) = -2x^4 + 9x^7 + x^3 + 10x^5$

L.C. _____ Degree: _____

4. $f(x) = 1 + 4x^2 - 3x - 2x^3$

L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

5. $f(-1) = 0, f(0) = 10,$ and $f(10) = 0.$

6. $f(-9) = 7, f(0) = 4,$ and $f(5) = 0.$

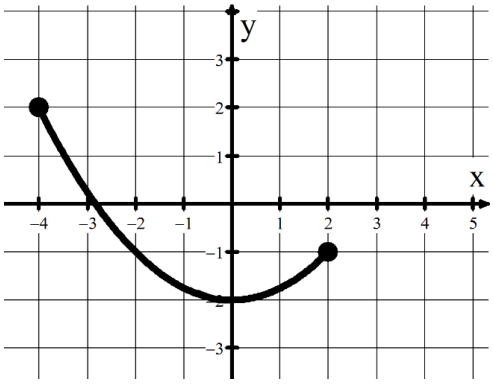
Is there a global maximum or minimum for each function?

7. $f(x) = -5x^8 + 2x^4 - 25$

8. $f(x) = 14x^2 + 1$

9. $f(x) = -2x^{11} - 5x^6 + x^2$

Find the following extrema. If there are none, cross it off and write NONE.



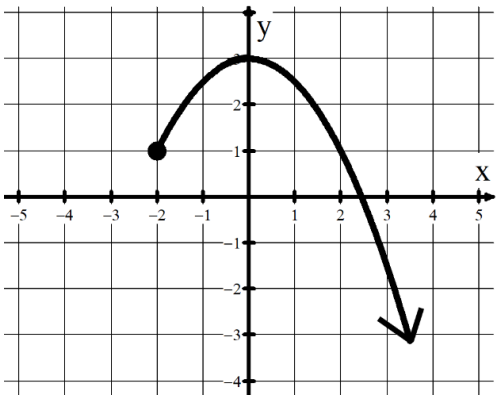
10. Absolute **min** of _____ when $x =$

11. Absolute **max** of _____ when $x =$

12. Relative **min**(s) at $x =$

13. Relative **max**(es) at $x =$

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14. Absolute **min** of _____ when $x =$

15. Absolute **max** of _____ when $x =$

16. Relative **min**(s) at $x =$

17. Relative **max**(es) at $x =$

Answers to 1.4 CA #2

1a. 3 1b. 1	2a. -3 2b. 4	3a. 9 3b. 7	4a. -2 4b. 3	5. Yes, on $-1 < x < 10$.
6. No guarantee.	7. max	8. min	9. none	10. min of -2 when $x = 0$
11. max of 2 when $x = -4$	12. 0	13. $-4,\ 2$	14. none	
15. max of 3 when $x = 0$	16. -2	17. 0		

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AP Precalc

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1.5A Polynomial Functions and Complex Zeros

1.5A Notes

Today, we will examine **complex zeros** of a polynomial function. *Complex* refers to both **real** numbers and **non-real** numbers.

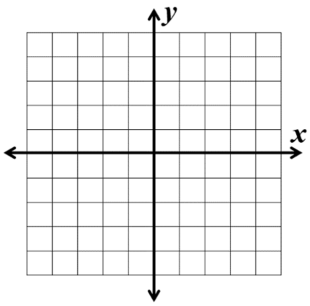
Zeros, x-intercepts, and Linear Factors

If we have $p(a) = 0$, then

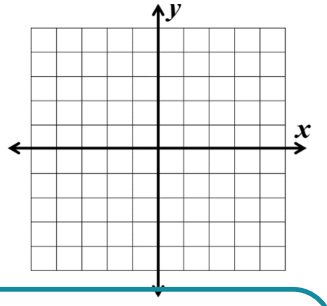
- …a is called a _____ of p.
- …we have an _____ at the point _____.
- …if a is a real number, then _____ is a _____ of p.

We are not reviewing all the factoring techniques from Algebra 2, but you will need to remember some basic factoring of quadratics. Here's an example.

1. Find all the x-intercepts of the function $g(x) = x^3 - 2x^2 - 8x$.



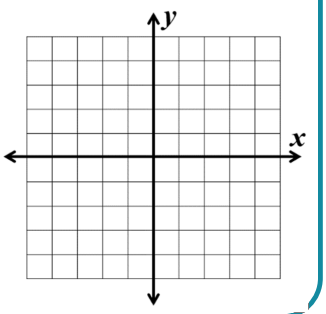
2. Given the polynomial function $h(x) = x^2 - 2x - 3$, what are all intervals on which $h(x) \geq 0$?



Multiplicity

If a linear factor $(x - a)$ is repeated n times, the corresponding zero of the polynomial function has a multiplicity n.

$$p(x) = (x + 1)^2(x - 3)^3(x + 4)$$



If the real zero, a, has even multiplicity, then the graph will "bounce" off the x-axis at $x = a$.

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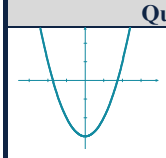
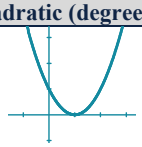
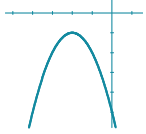
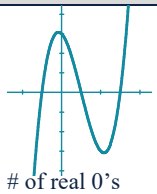
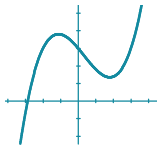
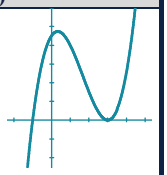
Another way of thinking about this is to look at the output values for input values near $x = a$. The output values will all have the same sign (positive or negative).

3. Given the polynomial function $p(x) = (x-2)(x-5)^4(x+7)$, what are all intervals on which $p(x) \leq 0$?

Complex Zeros

A polynomial of degree n has exactly n complex zeros when counting multiplicities. "Complex" refers to both real and non-real zeros.

If there are any non-real zeros, they always come in conjugate pairs (see explanation below). This means there are either no non-real zeros, or an even amount of non-real zeros.

Quadratic (degree = 2)			Cubic (degree = 3)		
					
# of real 0's	# of real 0's	# of real 0's	# of real 0's	# of real 0's	# of real 0's
# of non-real 0's	# of non-real 0's	# of non-real 0's	# of non-real 0's	# of non-real 0's	# of non-real 0's

4. The degree of a polynomial is 8 with real zeros at $x = -10, x = 5$, and $x = 16$. $x = 5$ has a multiplicity of 2. How many non-real zeros does the polynomial have?

Non-real Zeros

If $a + bi$ is a non-real zero of a polynomial p, then its conjugate _____ is also a zero of p.

Given one non-real zero of a polynomial, find another zero.	
5. $-3 + 6i$	6. $4 - 2i$

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Successive differences to find the degree of a polynomial.

If you have input and output values of a polynomial function, it is possible to find the degree of the function. This technique only works if the input values are over equal intervals. The degree of the polynomial function is equal to the least value n for which the successive nth differences are constant.

7. Find the degree of the polynomial from the given input and output values.

Input	0	1	2	3	4	5	6	7
Output	0	1	4	9	16	25	36	49

8. Find the degree of the polynomial from the given input and output values.

Input	0	1	2	3	4	5	6	7
Output	50	49	38	5	-62	-175	-346	-587

1.5A Polynomial Functions and Complex Zeros

AP Precalculus

1.5A Practice

For each polynomial function, find the intervals for each condition.

1. $f(x) = x^2 - 5x + 4$. When is $f(x) \leq 0$?

2. $g(x) = x^2 + 17x + 70$. When is $g(x) \geq 0$?

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3. $p(x) = (x-7)(x-1)^2$. When is $p(x) \leq 0$?

4. $h(x) = x^3 + 9x^2 + 18x$. When is $h(x) \geq 0$?

5. $a(x) = x(x-8)^3(x+3)^4$. When is $a(x) \geq 0$?

6. $f(x) = -x(x+4)^2(x+1)(x-6)^6$. When is $f(x) \leq 0$?

For each polynomial, the degree is listed along with all of its real zeros. Find the number of NON-REAL zeros the polynomial has.

7. The degree is 5 with real zeros at $x = -5, x = 1$, and $x = 4$.

8. The degree is 6 with real zeros at $x = -12$ and $x = 7$.

9. The degree is 8 with real zeros at $x = 0, x = 2$, and $x = 3$. $x = 2$ has a multiplicity of 4.

10. The degree is 16 with real zeros at $x = -2$ and $x = 10$. $x = -2$ has a multiplicity of 5.

11. The degree is 12 with real zeros at $x = 14, x = -6$, and $x = -10$. $x = 14$ has a multiplicity of 6.

12. The degree is 50 with real zeros at $x = 7$ and $x = 8$. $x = 8$ has a multiplicity of 19.

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Given one non-real zero of a polynomial, find another zero.

13. $7+2i$	14. $-5+i$	15. $1-5i$	16. $-3-4i$

Find the degree of the polynomial from the given input and output values.

17.

Input	0	1	2	3	4	5	6	7
Output	-2	-32	-162	-512	-1250	-2592	-4802	-8192

18.

Input	0	1	2	3	4	5	6	7
Output	6	24	54	96	150	216	294	384

19.

Input	0	1	2	3	4	5	6	7
Output	4	1	6	127	688	2,349	6,226	14,011

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20.

Input	0	1	2	3	4	5	6	7
Output	-10	-6	8	44	114	230	404	648

1.5A Polynomial Functions and Complex Zeros

1.5A Test Prep

21. A polynomial function has 3 real zeros and 4 non-real zeros. One of the real zeros has a multiplicity of 6. What is the degree of the polynomial?

(A) 7 (B) 9 (C) 12 (D) 13

---

22. **No calculator allowed!** The polynomial function g is given by $g(x) = (x-6)(x^2+2x+2)$. Which of the following describes the zeros of g?

(A) g has exactly two distinct real zeros.

(B) g has exactly three distinct real zeros.

(C) g has exactly one distinct real zero and no non-real zeros.

(D) g has exactly one distinct real zero and two non-real zeros.

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1.5A Polynomial Functions and Complex Zeros

CA #1

AP Precalculus Name: _____

For each polynomial function, find the intervals for each condition.

1. $h(x) = -x^2 + 8x - 7$. When is $h(x) \leq 0$?

2. $p(x) = x(x-2)^4(x-4)^3(x+8)$. When is $p(x) \leq 0$?

3. $f(x) = x^2(x+6)(x-1)^6(x+3)^3$. When is $f(x) \geq 0$?

4. $g(x) = x^3 - x^2 - 12x$. When is $g(x) \geq 0$?

For each polynomial, the degree is listed along with all of its real zeros. Find the number of NON-REAL zeros the polynomial has.

5. The degree is 4 with real zeros at $x = -4$ and $x = 4$.

6. The degree is 13 with real zeros at $x = -9, x = 0$, and $x = 13$. $x = 0$ has a multiplicity of 5.

7. The degree is 5 with real zeros at $x = 3$ and $x = 8$. $x = 3$ has a multiplicity of 4.

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Given one non-real zero of a polynomial, find another zero.

8. $-9+2i$

9. $3-7i$

Find the degree of the polynomial from the given input and output values.

10.

Input	0	1	2	3	4	5	6	7
Output	7	3	5	31	99	227	433	735

11.

Input	0	1	2	3	4	5	6	7
Output	8	17	58	113	116	-47	-538	-1567

Answers to 1.5A CA #1

1. $(-\infty,1]\cup[7,\infty)$	2. $(-\infty,-8]\cup[0,4]$	3. $(-\infty,-6]\cup[-3,\infty)$	4. $[-3,0]\cup[4,\infty)$

5. 2	6. 6	7. 0	8. $-9-2i$	9. $3+7i$	10. 3	11. 4

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1.5A Polynomial Functions and Complex Zeros

CA #2

AP Precalculus Name: _____

For each polynomial function, find the intervals for each condition.

1. $a(x) = -x^2 - x + 2$. When is $h(x) \geq 0$?

2. $f(x) = x^3 - 2x^2 - 35x$. When is $h(x) \leq 0$?

3. $w(x) = (x-3)(x-5)^3(x+7)^2$. When is $w(x) \leq 0$?

4. $p(x) = -x(x-9)^4(x-1)^2(x+4)^4$. When is $p(x) \geq 0$?

For each polynomial, the degree is listed along with all of its real zeros. Find the number of NON-REAL zeros the polynomial has.

5. The degree is 3 with real zeros at $x = -3$ and $x = 1$. $x = 1$ has a multiplicity of 2.

6. The degree is 9 with real zeros at $x = 2, x = 5$, and $x = 7$. $x = 7$ has a multiplicity of 3.

7. The degree is 22 with real zeros at $x = -6$ and $x = 8$. $x = -6$ has a multiplicity of 11

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Given one non-real zero of a polynomial, find another zero.

8. $4 - i$

9. $-3 + 6i$

Find the degree of the polynomial from the given input and output values.

10.

Input	0	1	2	3	4	5	6	7
Output	6	6	-12	-54	-102	-114	-24	258

11.

Input	0	1	2	3	4	5	6	7
Output	11	13	13	11	7	1	-7	-17

Answers to 1.5A CA #2

1. $[-2, 1]$	2. $(-\infty, -5] \cup [0,7]$	3. $[3, 5]$ and $x = -7$	4. $(-\infty, 0]$, $x = 1$ and $x = 9$

5. 0	6. 4	7. 10	8. $4 + i$	9. $-3 - 6i$	10. 4	11. 2

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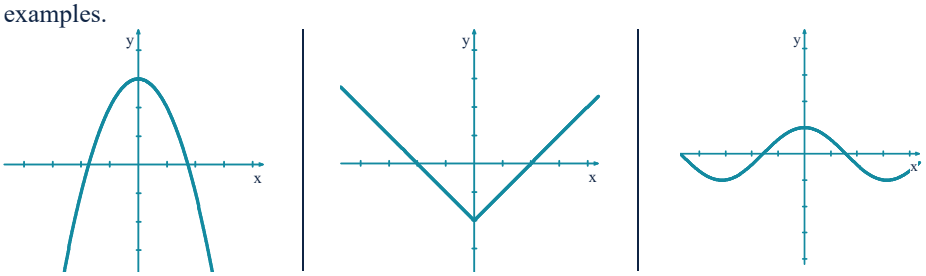
AP Precalc

Write your questions and thoughts here!

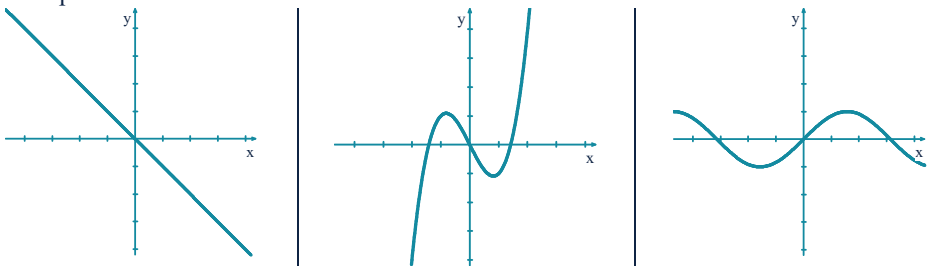
1.5B Even and Odd Polynomials

1.5B Notes

_____ functions are symmetrical over the _____. The following graphs are examples.

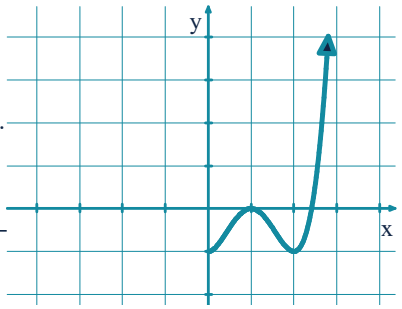


_____ functions are symmetrical over the _____. The following graphs are examples.



An EVEN Function…

- is graphically symmetric over the line $x = 0$.
- analytically has the property _____



You can prove that a function is even by substituting in $-x$ and see if you get the _____ original function.

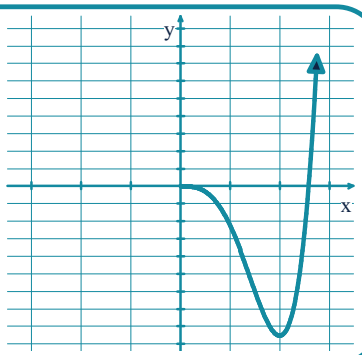
1. Show analytically that $f(x) = x^6 - 4x^2$ is an even function.

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Write your questions and thoughts here!

An ODD Function…

- is graphically symmetric over the point $(0,0)$.
- analytically has the property _____



You can prove that a function is odd by substituting in $-x$ and see if you get the _____ of the original function.

2. Show analytically that $f(x) = -2x^3 + 5x$ is an odd function.

3. Show analytically if $f(x) = 6x^4 - 2x$ is even, odd, or neither.

Shortcut for simple polynomial form

Let p be a polynomial of the form $p(x) = a_n x^n$, where $n \geq 1$ and $a_n \neq 0$.

- If n is even, then p is an even function.
- If n is odd, then p is an odd function.

If you see one of these in the practice, you can just state that it is an odd or even function without having to go through all the analytical work!

State if the following functions are odd or even.		
4. $f(x) = -10x^3$	5. $f(x) = 7x^4$	6. $f(x) = -3x^6$

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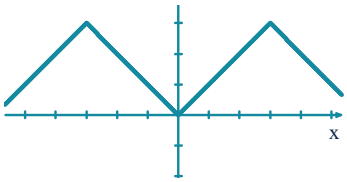
1.5B Even and Odd Polynomials

AP Precalculus

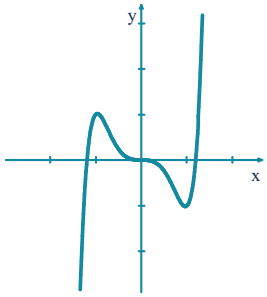
1.5B Practice

State whether the following graphs represent functions that are even, odd, or neither.

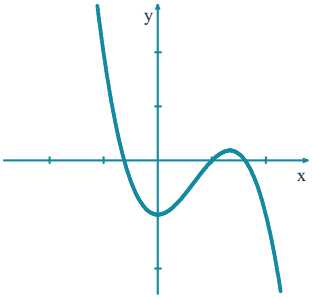
1.



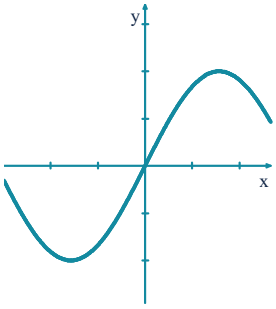
2.



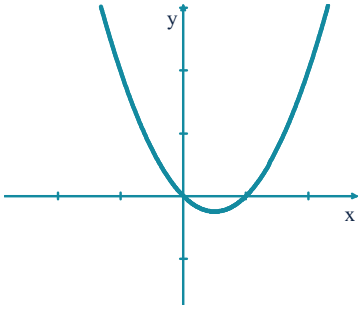
3.



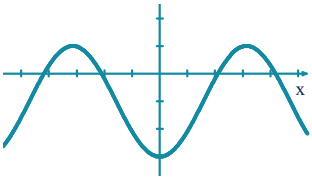
4.



5.



6.



State if the following functions are even, odd, or neither.

7. $f(x) = x^3 + x$

8. $g(x) = x^4 + 3x^2$

9. $h(x) = -7x^6$

10. $w(t) = 8t^5 + 2t$

11. $a(w) = 1 - w - w^3$

12. $f(x) = 2x^{10} + x^6 - 7x^2 - 2$

13. $h(t) = 4t^9$

14. $g(x) = x^2 + x - 3$

15. $b(x) = -4x^7 + x^3 - x$

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16. Is $f(x) = 7x^6 - x^4 + 5$ even, odd, or neither

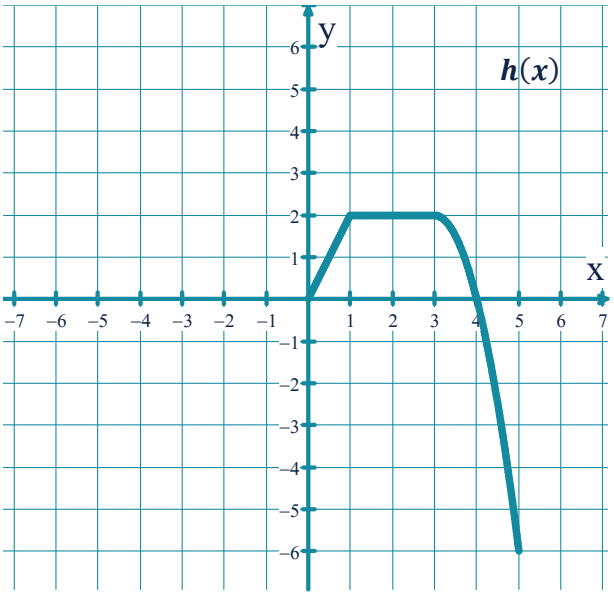
1.5B Even and Odd Polynomials

1.5B Test Prep

17. Certain values of $g(x)$ are given in the table below. Given that $g(x)$ is odd, fill in the missing values of the table.

x	$g(x)$
-5	10.5
7	
	-38.5
-7	22.5
5	
9	38.5

18. Given that $h(x)$ is continuous on $-5 \le x \le 5$ and odd, draw the graph $h(x)$ from $-5 \le x \le 0$.



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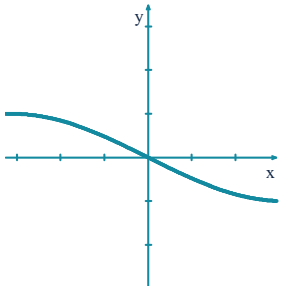
1.5B Even and Odd Polynomials

CA #1

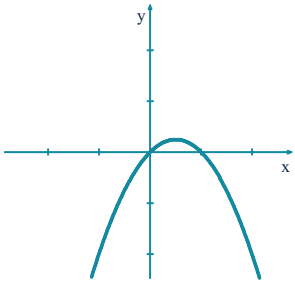
AP Precalculus Name: _____

State whether the following graphs represent functions that are even, odd, or neither.

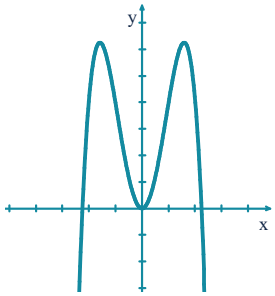
1.



2.



3.



State if the following functions are even, odd, or neither.

4. $f(x) = x^3 + 3x$

5. $f(x) = 3x^6 - 5x^4 - 2x^2$

6. $f(x) = 2x^7 - x^5 + x$

7. $y = x^2 + 3x - 5$

8. $f(x) = \frac{1}{2}x^2 + 5$

Answers to 1.5B CA #1

1. odd	2. neither	3. even	4. odd, $f(-x) = -f(x)$
5. even, $f(-x) = f(x)$	6. odd, $f(-x) = -f(x)$	7. neither	8. even, $f(-x) = f(x)$

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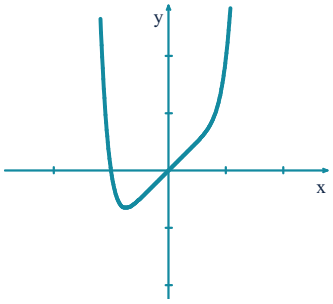
1.5B Even and Odd Polynomials

CA #2

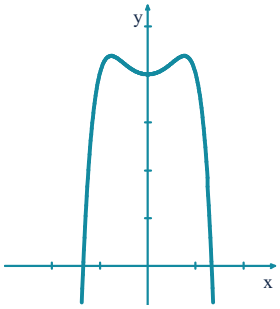
AP Precalculus Name: _____

State whether the following graphs represent functions that are even, odd, or neither.

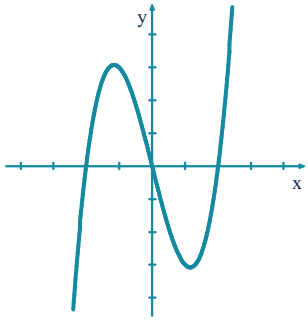
1.



2.



3.



State if the following functions are even, odd, or neither.

4. $f(x) = x^3 - 2x^5$

5. $f(x) = 1 - 2x^5$

6. $f(x) = x^{10} - 7x^2 + 3$

7. $f(x) = 6x^9 + x^3 - 2x$

8. $f(x) = 6 - 3x^2 + x^4$

Answers to 1.5B CA #2

1. neither	2. even, $f(-x) = f(x)$	3. odd, $f(-x) = -f(x)$	4. odd, $f(-x) = -f(x)$
5. neither	6. even, $f(-x) = f(x)$	7. odd, $f(-x) = -f(x)$	8. even, $f(-x) = f(x)$

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AP Precalc

Write your questions and thoughts here!

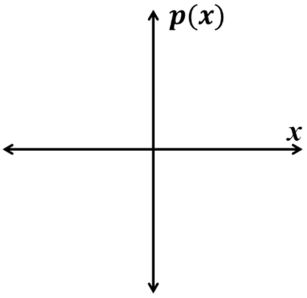
1.6 Polynomial Functions and End Behavior

1.6 Notes

Today's lesson focuses on the shape of polynomial graphs and how to describe their end behaviors.

The "left" and "right" side of a polynomial function graph will either go "up" or "down".

The Left Side
The x values are approaching negative infinity. The notation is:

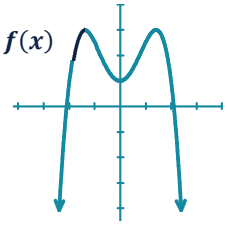


The Right Side
The x values are approaching positive infinity. The notation is:

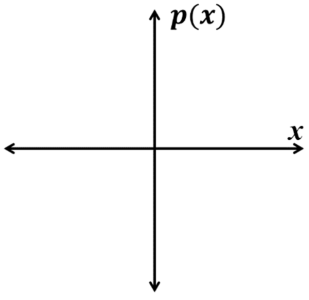
As the x-values approach negative infinity or positive infinity, do the y-values get larger or smaller? (Does the left side of the graph go up or down?) The following chart is how we describe the end behavior with fancy mathematical notation.

	Left side	Right side
Up		
Down		

1. Use the graph of $f(x)$, shown to the right, and describe its end behavior using limit notation.



2. Using the axis to the right, draw an example of a polynomial function that matches the statements $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$.



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Write your questions and thoughts here!

You don't need a graph to describe the end behaviors of a polynomial function. You only need to recognize the _____.

The leading term contains the variable with the largest exponent. As $x \to \infty$ or $x \to -\infty$, this leading term dominates all the other smaller terms. We don't need to worry about the smaller terms. In other words…

$$p(x) = 7x^6 - 4x^5 + 8x^4 + x^3 - 3x^2 - 9x + 10$$

If the leading term has an _____ degree, the left and right side will behave the _____.

	Left side $x \to -\infty$	Right side $x \to \infty$
EVEN degree and **positive** leading coefficient $p(x) = x^2$		
EVEN degree and **negative** leading coefficient $p(x) = -x^2$		

If the leading term has an _____ degree, the left and right side will behave the _____.

	Left side $x \to -\infty$	Right side $x \to \infty$
ODD degree and **positive** leading coefficient $p(x) = x^3$		
ODD degree and **negative** leading coefficient $p(x) = -x^3$		

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Write your questions and thoughts here!

Describe the end behavior of each polynomial function using limit notation.

3. $p(x) = -x^3 - x^2 + x$

4. $p(x) = 6x^5 - x^4 + 1$

5. $p(x) = -3x^4 + x^3 + x - 2$

1.6 Polynomial Functions and End Behavior

AP Precalculus

1.6 Practice

Describe the end behavior of each function using limit notation.

1. $g(x) = 4x^7 - 3x^4 + x$

2. $p(x) = 7x^4 + 3x^3 - 3x - 4$

3. $f(x) = -7x^9 - 8x^3 + 6$

4. $g(x) = -3x^6 + 5x^3 - 2x + 6$

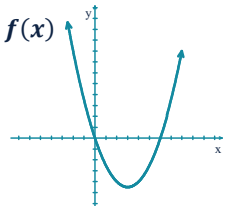
5. $f(x) = x^3 + 4x^2 - 3$

6. $p(x) = -8x^2 - 3x + 10$

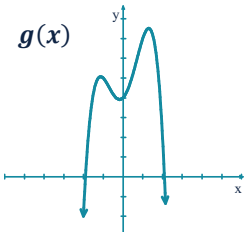
7. $p(x) = -6x^5 + x^4 + 5x^2 - 4$

8. $f(x) = 3x^2 + 8x + 11$

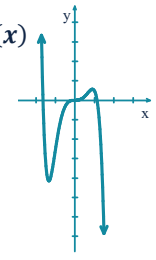
9. $f(x)$



10. $g(x)$



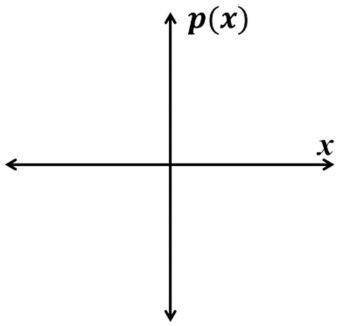
11. $h(x)$



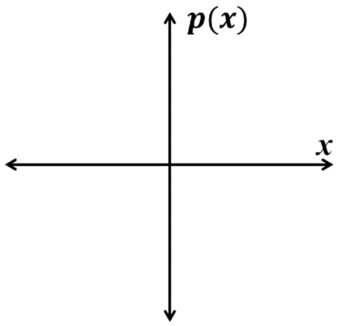
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Sketch the graph of a polynomial function that could match each statement.

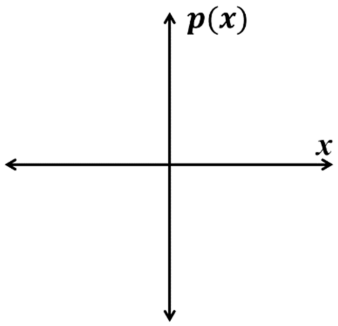
12. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = \infty$



13. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$



14. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$



1.6 Polynomial Functions and End Behavior

1.6 Test Prep

15. The following polynomial function f is given by $f(x) = -7x^6 + 2x^2 + 4$. Which of the following statements about the end behavior of f is true?

(A) The sign of the leading term of f is positive, and the degree of the leading term of f is even; therefore, $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$.

(B) The sign of the leading term of f is negative, and the degree of the leading term of f is odd; therefore, $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = -\infty$.

(C) The sign of the leading term of f is positive, and the degree of the leading term of f is odd; therefore, $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$.

(D) The sign of the leading term of f is negative, and the degree of the leading term of f is even; therefore, $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = -\infty$.

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1.6 Polynomial Functions and End Behavior

CA #1

AP Precalculus Name: _____

Describe the end behavior of each function using limit notation.

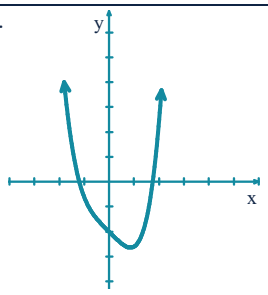
1. $g(x) = -5x^7 + 6x^2$

2. $p(x) = -7x^2 + 1$

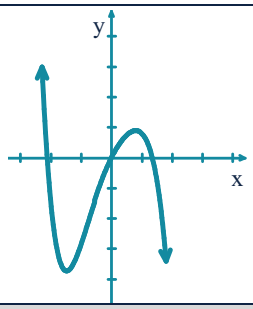
3. $f(x) = 3x^6 - x^3 + 2x$

4. $f(x) = 6x^5 - 3x^2 - 7$

5.

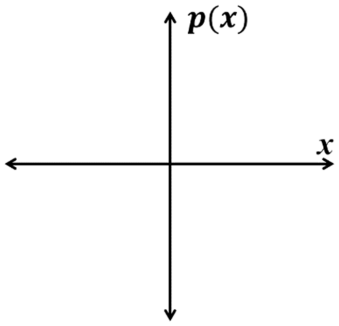


6.

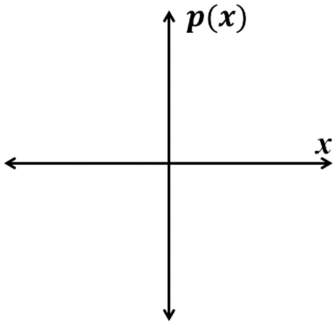


Sketch the graph of a polynomial function that could match each statement.

7. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$

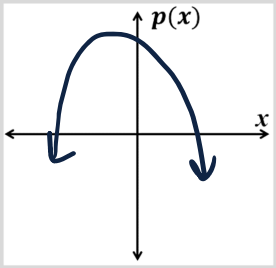
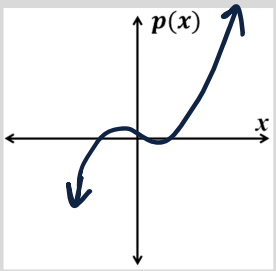


8. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = \infty$



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Answers to 1.6 CA #1

1. $\lim_{x\to-\infty} g(x) = \infty$ and $\lim_{x\to\infty} g(x) = -\infty$	2. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$
3. $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$	4. $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$
5. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = \infty$	6. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$
7. 	8. 

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1.6 Polynomial Functions and End Behavior

CA #2

AP Precalculus Name: _____

Describe the end behavior of each function using limit notation.

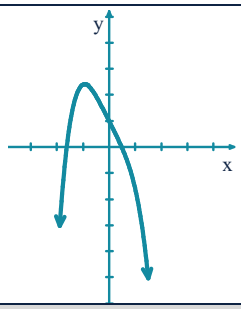
1. $p(x) = -3x^{10} + 9$

2. $f(x) = 12x^7 - 6x^2 + 4x - 8$

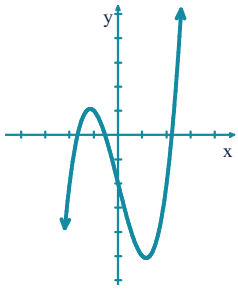
3. $p(x) = 5x^8 - 6x^4 + 7$

4. $g(x) = -9x^3 + 2$

5.

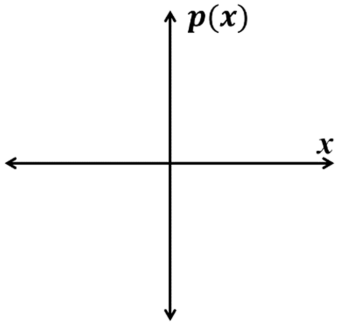


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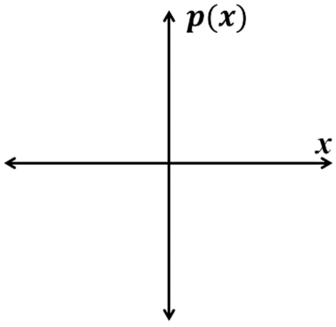


Sketch the graph of a polynomial function that could match each statement.

7. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = \infty$

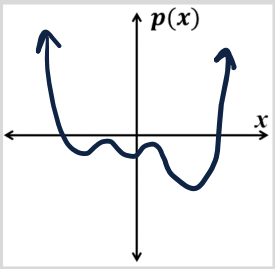
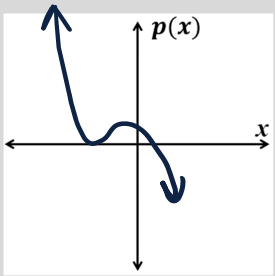


8. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$



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Answers to 1.6 CA #2

1. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$	2. $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$
3. $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$	4. $\lim_{x\to-\infty} g(x) = \infty$ and $\lim_{x\to\infty} g(x) = -\infty$
5. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$	6. $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = \infty$
7. 	8. 

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AP Precalculus

Name: _____ Date:_____ Period: _____ **1A Review**

Unit 1A Review – Polynomial and Rational Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 1.1 – 1.6.

1. For the function $s(t)$, s is the number of people swimming at the local pool and t is the temperature measured in Fahrenheit. Identify the dependent and independent variables.

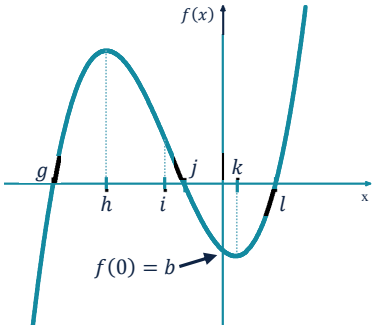
   Dep:

   Indep:

2. Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the interval $9 < x < 17$ and justify your answer.

x	9	11	13	15	17
$f(x)$	8	12	15	17	18

3. Use the graph to the right to answer the questions below.
   a. On what interval(s) is the graph concave up?
   b. On what interval(s) is the graph concave down?
   c. On what interval(s) is the graph increasing?
   d. On what interval(s) is the graph decreasing?
   e. Find the zeros of the function.
   f. Find the y-intercept of the function.



4. **Calculator active.** Find the average rate of change of the function $w(r) = \sqrt{1 - 4r}$ on the interval $-5 \le r \le -1$.

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5. **Calculator active.** Using the information in the table below, find the average rate of change for each given interval. Include units.

t seconds	24	60	135	200
$s(t)$ miles	8	1.3	3.5	0.3

a. $135 \leq t \leq 200$

b. $24 \leq t \leq 135$

c. $24 \leq t \leq 200$

6. **Calculator active.** Estimate the rate of change of $f(x) = x^2 - x$ at $x = -1$

7. Mr. Gardener is decreasing the amount of water used on his lawn each month, and the height of his grass is decreasing. Does this scenario represent a positive or negative rate of change?

What is the average rate of change for each function on the given intervals?

8. $y = 7 - 2x$ on $-4 \leq x \leq 1$

9. $y = 3x^2 - 2x + 1$ on $-1 \leq x \leq 2$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

10. $y = 5x + 9$

11. $f(x) = 2x - 5x^2$.

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12. The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

x	5	9	13	17	21
$g(x)$	45	20	0	-10	-14

Find the leading coefficient and the degree of each polynomial.

13. $f(x) = x^5 - 2x^2$

L.C. _____ Degree: _____

14. $f(x) = 10 - 3x^2 + 7x^3 - 2x$

L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

15. $f(0) = -3, f(4) = 0,$ and $f(7) = 0.$

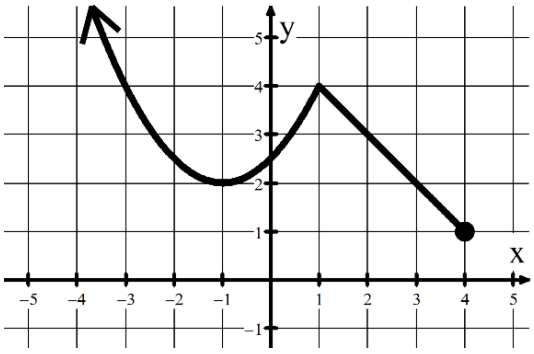
16. $f(-7) = 0, f(-2) = 5, f(0) = 1,$ and $f(9) = 0.$

Is there a global maximum or minimum for each function?

17. $f(x) = -5x^6 + 6x^4 - 3x^3 + 1$

18. $f(x) = 2x^5 + x^2 - 6$

Find the following extrema. If there are none, cross it off and write NONE.



19. Absolute **min** of _____ when $x =$

20. Absolute **max** of _____ when $x =$

21. Relative **min**(s) at $x =$

22. Relative **max**(es) at $x =$

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For each polynomial function, find the intervals for each condition.

23. $h(x) = x^2 - 8x + 15$. When is $h(x) \leq 0$?

24. $f(x) = -x^3 + 5x^2 + 24x$. When is $f(x) \geq 0$?

25. The degree of a polynomial is 7 with real zeros at $x = -8, x = 1$, and $x = 4$. $x = 1$ has a multiplicity of 3. How many non-real zeros does the polynomial have?

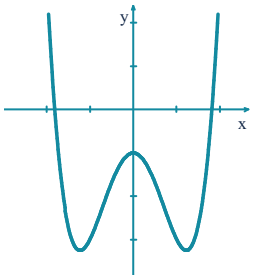
26. $5 - i$ is a non-real zero of a polynomial, find another zero.

27. Find the **degree** of the polynomial from the given input and output values.

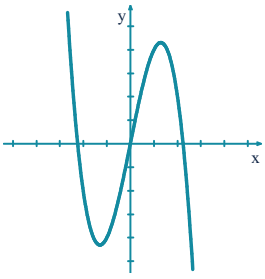
Input	0	1	2	3	4	5	6	7
Output	2	-2	4	20	46	82	128	184

State whether the following graphs represent functions that are even, odd, or neither.

28.



29.



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State if the following functions are even, odd, or neither.

30. $f(x) = 4x^7 + 5x^3 - 2x$

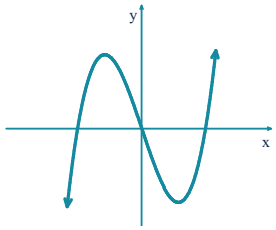
31. $f(x) = 7 - 6x^8 - 3x^2$

Describe the end behavior of each function using limit notation.

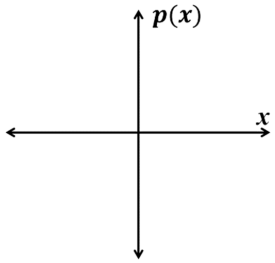
32. $p(x) = -11x^7 - 6x^2 + 4x$

33. $p(x) = -7x^6 + 4x - 8$

34.



35. Sketch the graph of a polynomial function that could match statements $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = \infty$.



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AP Precalculus

Name: _____ Date:_____

Corrective Assignment

Unit 1A CA – Polynomial and Rational Functions

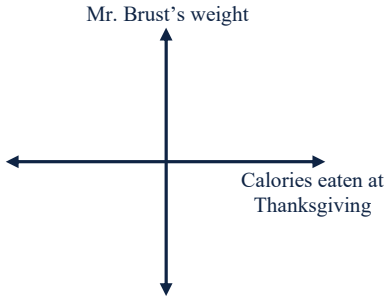
For each function, identify what the dependent and independent variables represent.

1. $r(c)$ is a function where c is the number of coyotes and r is the number of rabbits on the property.

Dep:

Indep:

2.



Dep:

Indep:

3. Let the function f be increasing or decreasing, but not both. State whether the function is increasing or decreasing on the given interval and justify your answer.

x	15	30	50	75	110
$f(x)$	43	40	35	25	12

4.

a. On what interval(s) is the graph concave up?

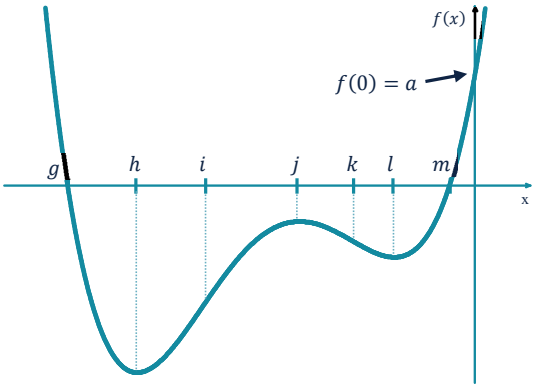
b. On what interval(s) is the graph concave down?

c. On what interval(s) is the graph increasing?

d. On what interval(s) is the graph decreasing?

e. Find the zeros of the function.

f. Find the y-intercept of the function.



5. **Calculator active.** Find the average rate of change of the function $h(t) = t + 3^t$ on the interval $\frac{1}{2} \le x \le 1$.

6. Using the information in the table below, find the average rate of change for each interval.

t weeks	2	4	6	8
$B(t)$ wild boar	2	12	17	20

a. $2 \le t \le 8$

b. $6 \le t \le 8$

c. $2 \le t \le 6$

7. **Calculator active.** Estimate the rate of change of $f(x) = \frac{1}{x}$ at $x = 2$.

8. Increasing the number of cigarettes someone smokes increases their chance of lung cancer. Does this scenario represent a positive or negative rate of change?

What is the average rate of change for each function on the given intervals?

9. $y = 3x - 6$ on $2 \le x \le 5$

10. $y = -x^2 + 3x + 4$ on $-2 \le x \le 1$

What is the rate of change of the average rates of change for each function over consecutive equal-length intervals?

11. $y = 7 - 3x$

12. $f(x) = 2x^2 + 3x - 6.$

13. The values of a function are given at selected x-values in the table below. The function's concavity does not change. Determine if the function is concave up or concave down. Justify your answer.

x	6	8	10	12	14
$g(x)$	8	6	0	-14	-34

Find the leading coefficient and the degree of each polynomial.

14. $f(x) = 4x^2 - 3$

L.C. _____ Degree: _____

15. $f(x) = -8x^3 + 5 + 9x^4$

L.C. _____ Degree: _____

Let $f(x)$ be a polynomial function with the given values. Are there any guaranteed extrema? If so, state where they occur.

16. $f(-13) = 0, f(0) = 5,$ and $f(2) = 3.$

17. $f(-1) = 0, f(0) = -8, f(4) = -3,$ and $f(6) = 0.$

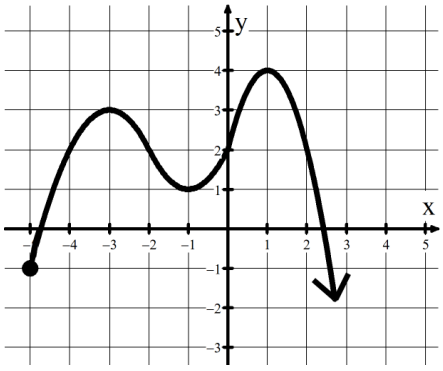
Is there a global maximum or minimum for each function?

18. $f(x) = -2x^7 + 1$

19. $f(x) = 3x^4 - 4x^3 - 2$

20. $f(x) = -2x^6 + 7x^5 - x^3 + x$

Find the following extrema. If there are none, cross it off and write NONE.



21. Absolute **min** of _____ when $x =$

22. Absolute **max** of _____ when $x =$

23. Relative **min**(s) at $x =$

24. Relative **max**(es) at $x =$

For each polynomial function, find the intervals for each condition.

25. $f(x) = -x^2 - 4x + 45$. When is $f(x) \geq 0$?

26. $g(x) = (x+5)^2(x-1)^3(x-6)$. When is $g(x) \geq 0$?

For each polynomial, the degree is listed along with all of its real zeros. Find the number of NON-REAL zeros the polynomial has.

27. The degree of a polynomial is 5 with real zeros at $x = -3, x = 0$, and $x = 2$. $x = 2$ has a multiplicity of 3.

28. The degree of a polynomial is 10 with real zeros at $x = -5, x = -2$, and $x = 11$. $x = 11$ has a multiplicity of 4.

Given one non-real zero of a polynomial, find another zero.

29. $9 + 2i$

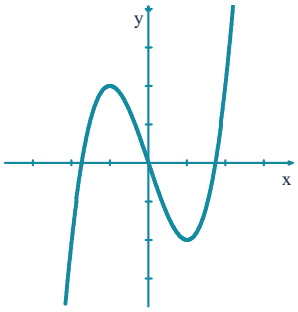
30. $-8 - i$

31. Find the **degree** of the polynomial from the given input and output values.

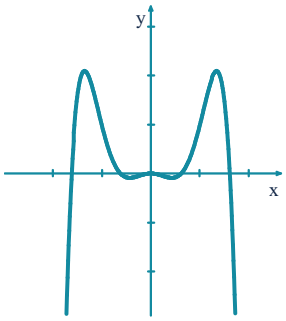
Input	0	1	2	3	4	5	6	7
Output	100	107	118	115	80	-5	-158	-397

State whether the following graphs represent functions that are even, odd, or neither.

32.



33.



State if the following functions are even, odd, or neither.

34. $f(x) = x^4 + x + 1$

35. $f(x) = 6x^5 - 2x^3 + x$

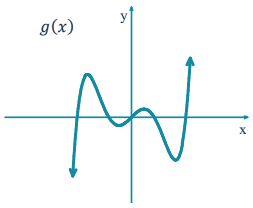
36. $f(x) = 3 + 5x^6 - 3x^2$

Describe the end behavior of each function using limit notation.

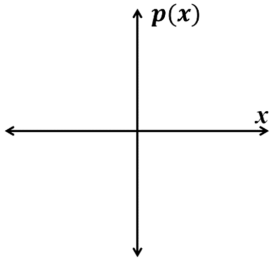
37. $f(x) = 7x^4 - 12$

38. $p(x) = -8x^{13} - 6x^4 + x$

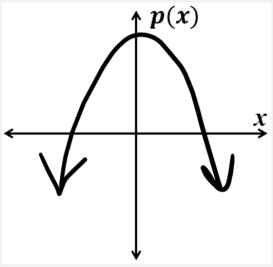
39.



40. Sketch the graph of a polynomial function that could match statements $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$.



Answers to Unit 1A Corrective Assignment

1. Dependent: Number of rabbits. Independent: Number of coyotes.			2. Dependent: Mr. Brust's weight. Independent: Calories eaten at Thanksgiving.		
3. f is decreasing on the interval $15 < x < 110$ because for all a and b in the interval, if $a < b$, then $f(a) > f(b)$.			4a. $x < i$ and $x > k$	4b. $i < x < k$	
4c. $h < x < j$ and $x > l$	4d. $x < h$ and $j < x < l$		4e. $x = g$ and $x = m$	4f. $y = a$	
5. 3.5358	6a. 3 wild boar per week	6b. 1.5 wild boar per week	6c. 3.75 wild boar per week		
7. ≈ -0.25	8. positive	9. 3	10. 4	11. 0	12. 4
13. Concave down because the rate of change is decreasing over equal-length input-value intervals.			14a. 4	14b. 2	15a. 9	15b. 4
16. No guarantee.	17. Yes, on $-1 < x < 6$.		18. none	19. min	20. max
21. none	22. max of 4 when $x = 1$		23. -5 and -1	24. -3 and 1	
25. $[-9, 5]$	26. $(-\infty, 1] \cup [6, \infty)$	27. 0	28. 4	29. $9 - 2i$	30. $-8 + i$
31. 3	32. odd	33. even	34. Neither	35. Odd, $f(-x) = -f(x)$	
36. Even, $f(-x) = f(x)$	37. $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$		38. $\lim_{x\to-\infty} p(x) = \infty$ and $\lim_{x\to\infty} p(x) = -\infty$		
39. $\lim_{x\to-\infty} g(x) = -\infty$ and $\lim_{x\to\infty} g(x) = \infty$			40. 		

AP Precalc

Write your questions and thoughts here!

1.7A Rational Functions & End Behavior — 1.7A Notes

Rational Function

The ratio of two polynomials where the polynomial in the denominator cannot equal 0.

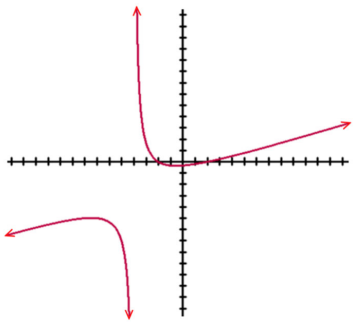
$r(x) = \frac{f(x)}{g(x)}$ where f and g are polynomials and $g(x) \neq 0$

Domain of Rational Functions

Example 1:

1. $f(x) = \frac{x^2-4}{3x+12}$

Domain:



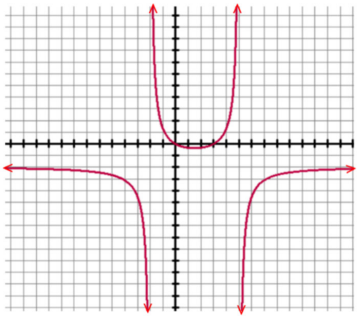
Find the domain of the rational functions.

2. $g(x) = \frac{4x-1}{x^2-2x-24}$

Domain:

3.

Domain:

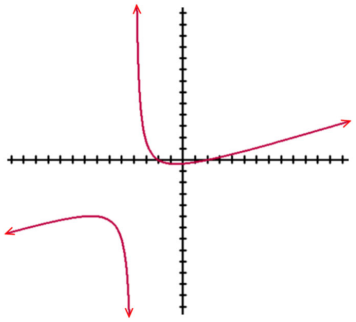


End Behavior of Rational Functions

Example 4:

4. $f(x) = \frac{x^2-4}{3x+12}$

End Behavior:



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Write your questions and thoughts here!

Determine the end behavior of the following rational functions.

5. $g(x) = \frac{4x-1}{x^2-2x-24}$

As x increases without bound the $g(x)$…

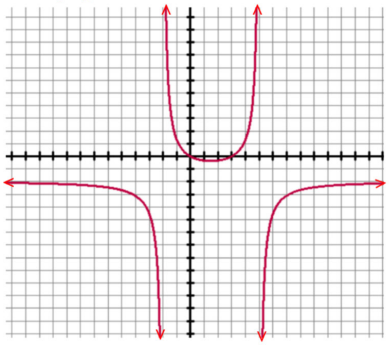
x	$g(x)$
10	
100	
1,000	
10,000	

As x decreases without bound the $g(x)$…

x	$g(x)$
-10	
-100	
$-1{,}000$	
$-10{,}000$	

End Behavior:

6. The graph of f



End Behavior:

Horizontal Asymptote

The line $y = b$ is a horizontal asymptote of the graph of f when…

$$\lim_{x\to-\infty} f(x) = b \qquad \text{OR} \qquad \lim_{x\to\infty} f(x) = b$$

1.7A Rational Functions and End Behavior

AP Precalculus

1.7A Practice

State the domain of the following rational functions. Use interval notation.

1. $f(x) = \frac{x+1}{2x+5}$

Domain:

2. $g(n) = \frac{n^2-4}{(n+3)(n-7)}$

Domain:

3. $h(x) = \frac{x^3+7x^2+12x}{x^2+2x-15}$

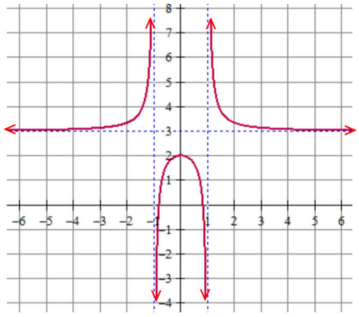
Domain:

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State the domain of the following rational functions. Use interval notation.

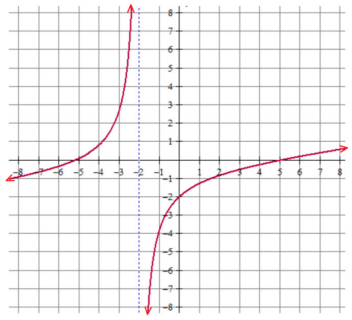
4.

Domain:



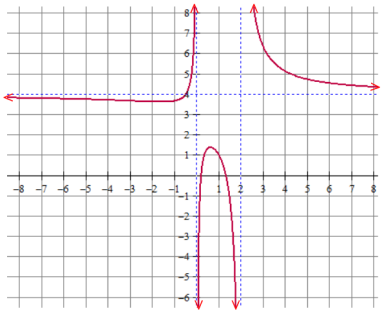
5.

Domain:



Use the graph of the rational function *f* to find the following.

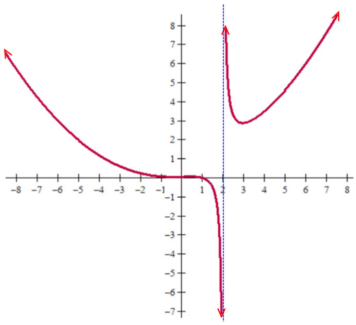
6.



End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

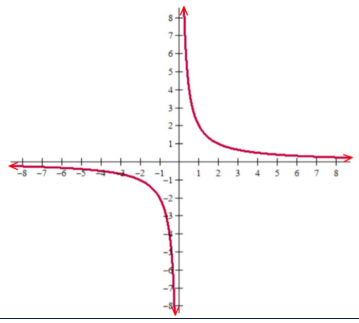
7.



End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

8.



End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

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CALCULATOR ACTIVE Complete the table to answer the following.

9. $f(x) = \frac{x^2}{x^2-4x}$

x	$-10{,}000$	$-1{,}000$	-100	100	$1{,}000$	$10{,}000$
$f(x)$						

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

10. $d(t) = \frac{t^3-3t}{2t+1}$

t	$-5{,}000$	-500	-50	50	500	$5{,}000$
$d(t)$						

End Behavior:

Is there a horizontal asymptote?
If so, write the equation of the horizontal asymptote.

Use the table of the rational function h to find the following.

11.

t	$h(t)$
$-999{,}999$	4.2×10^{32}
$-99{,}999$	7.3×10^{12}
$-9{,}999$	$6{,}524{,}378$
-999	$561{,}391$
0	32
999	-478
$9{,}999$	$-289{,}437$
$99{,}999$	$-2{,}198{,}235$
$999{,}999$	$-389{,}031{,}455$

a. Find $h(999) =$

b. Find $\lim_{t\to\infty} h(t) =$

c. As t increases without bound the $h(t)$…

d. Find the y-intercept.

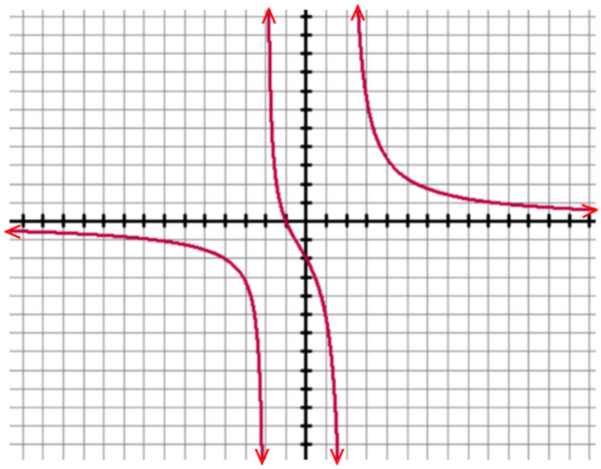
e. Find $\lim_{t\to-\infty} h(t) =$

f. As t decreases without bound the $h(t)$…

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Use the graph of the rational function f to answer the following if they exist!

12.



a. State the domain of $f(x)$.
Use interval notation.

b. Find $f(9) =$

c. Find the y-intercept.

d. Find the x-intercept(s).

e. State the interval(s) where $f(x)$ is increasing.

f. State the interval(s) where $f(x)$ is decreasing.

g. Find $\lim_{x\to\infty} f(x) =$

h. As x increases without bound the $f(x)$…

i. Estimate the point of inflection.

j. Find the average rate of change over the interval $-1 \le x \le 0$.

k. Find $\lim_{x\to-\infty} f(x) =$

l. As x decreases without bound the $f(x)$…

m. Find the horizontal asymptote.

n. State the interval(s) where $f(x) < 0$.

13. Which one of the following polynomials could be the denominator for the graph of the rational function shown above in question #12 ?

(A) $x^2 + 4x + 4$

(B) $x^2 - 4x + 4$

(C) $x^2 - 4$

(D) $(x - 4)^2$

(E) $(x + 4)^2$

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1.7A Rational Functions and End Behavior

1.7A Test Prep

Multiple Choice

14. Given $f(x) = x^2 + a^2$ and $g(x) = x^2 - a^2$ where a is a constant integer. The function $r(x) = \frac{f(x)}{g(x)}$. What is the domain of $r(x)$?

(A) $(-\infty, -a) \cup (a, \infty)$

(B) $(-a, a)$

(C) $(-\infty, -a) \cup (-a, a) \cup (a, \infty)$

(D) $(-\infty, -a)$

(E) (a, ∞)

15. The function $f(x)$ is defined such that for all $x > 0$, the line $y = 4$ is a horizontal asymptote. Which of the following must be true?

(A) $f(4)$ is undefined

(B) $f(x) \neq 4$ for all $x > 0$

(C) $\lim_{x \to \infty} f(x) = 0$

(D) $\lim_{x \to \infty} f(x) = 4$

(E) $f(4) = 0$

16. Given the line $y = a$ is a horizontal asymptote of $g(x)$ for all $x < 0$, which of the following must be true?

(A) $g(a)$ is undefined

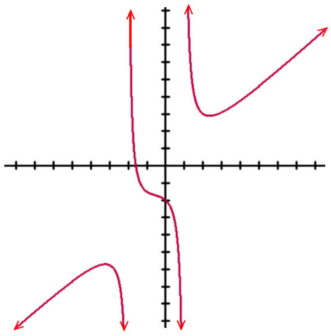
(B) $g(x) \neq a$

(C) $\lim_{x \to -\infty} g(x) = a$

(D) $\lim_{x \to -\infty} g(x) = -a$

(E) None of the above

17. Given the graph of f. Which describes the end behavior of f?



(A) $\lim_{x \to -\infty} f(x) = -\infty$ and $\lim_{x \to \infty} f(x) = -\infty$

(B) $\lim_{x \to -\infty} f(x) = \infty$ and $\lim_{x \to \infty} f(x) = \infty$

(C) $\lim_{x \to -\infty} f(x) = -\infty$ and $\lim_{x \to \infty} f(x) = \infty$

(D) $\lim_{x \to -\infty} f(x) = \infty$ and $\lim_{x \to \infty} f(x) = -\infty$

(E) $\lim_{x \to \infty} f(x) = f(0)$

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1.7A Rational Functions and End Behavior

AP Precalculus Name: _____ **CA #1**

State the domain of the following rational functions. Use interval notation.

1. $f(x) = \frac{x^3+1}{x^2-16}$

Domain:

2. $h(x) = \frac{x+3}{2x^2+10x-28}$

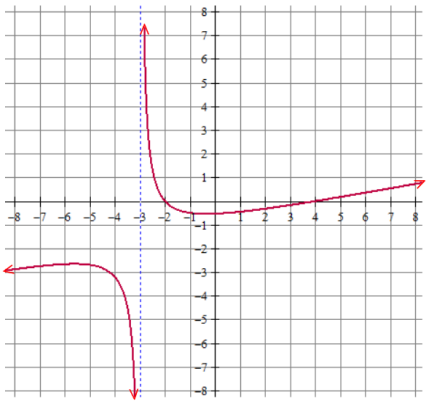
Domain:

3. $d(t) = \frac{t^2-3t+1}{4-3t}$

Domain:

Use the graph of the rational function f to find the following.

4.

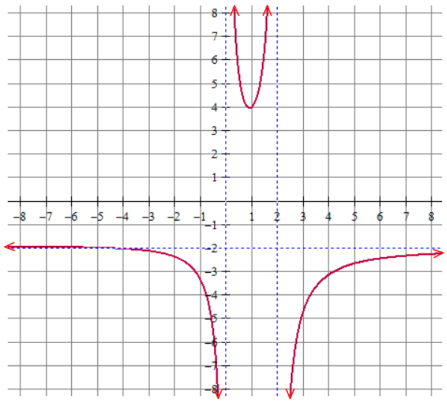


Domain:

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

5.



Domain:

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

Continued on the back.

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CALCULATOR ACTIVE Complete the table to answer the following.

6. $f(x) = \frac{x^2}{x^2-4x}$

x	$-10{,}000$	$-1{,}000$	-100	100	$1{,}000$	$10{,}000$
$f(x)$						

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

7. $d(t) = \frac{t^3-3t}{2t+1}$

t	$-5{,}000$	-500	-50	50	500	$5{,}000$
$d(t)$						

End Behavior:

Is there horizontal aysmptote?
If so, write the equation of the horizontal aysmptote.

Answers to 1.7A CA #1

1. $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$
2. $(-\infty, -7) \cup (-7, 2) \cup (2, \infty)$
3. $\left(-\infty, \frac{4}{3}\right) \cup \left(\frac{4}{3}, \infty\right)$
4. Domain: $(-\infty, -3) \cup (-3, \infty)$
   End Behavior:
   $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$
   Horizontal Asymptote: none
5. Domain: $(-\infty, 0) \cup (0, 2) \cup (2, \infty)$
   End Behavior:
   $\lim_{x\to-\infty} f(x) = -2$ and $\lim_{x\to\infty} f(x) = -2$
   Horizontal Asymptote: $y = -2$

6.

x	$f(x)$
-10,000	0.9996
-1,000	0.99602
-100	0.9615
100	1.0417
1,000	1.004
10,000	1.0004

$\lim_{x\to-\infty} f(x) = 1$
$\lim_{x\to\infty} f(x) = 1$
HA: $y = 1$

7.

t	$d(t)$
-5,000	$\approx 1.2 \times 10^7$
-500	125124
-50	1261.1
50	1236.1
500	124874
5,000	$\approx 1.2 \times 10^7$

$\lim_{t\to-\infty} f(x) = \infty$
$\lim_{t\to\infty} f(x) = \infty$
HA: none

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